

Positively curved Einstein four-manifolds

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Abstract

We prove the classification conjecture for connected smooth closed Einstein four-manifolds with strictly positive sectional curvature. Up to positive scaling and isometry, every such manifold is the round four-sphere, the complex projective plane with its Fubini–Study metric, or real projective four-space with its round metric. No orientability assumption is needed.

1 Introduction

A Riemannian metric is Einstein if its Ricci tensor is a constant multiple of the metric. This fixes the trace of the sectional curvatures while allowing individual curvatures to vary. The classification conjecture in dimension four asks whether the round S^4 , the round $\mathbb{RP}^4$, and Fubini–Study $\mathbb{CP}^2$ exhaust the connected closed Einstein four-manifolds with strictly positive sectional curvature, up to positive scaling and isometry. Yang explicitly lists these three models in Conjecture 1 of [32]. We prove the classification in the closed setting.

Theorem 1.1. *Let (M, g) be a connected smooth closed Einstein four-manifold with strictly positive sectional curvature, without an orientability assumption. There is a constant $a > 0$ such that (M, ag) is isometric to the standard round S^4 , to $\mathbb{CP}^2$ with its Fubini–Study metric, or to $\mathbb{RP}^4$ with its standard round metric.*

Strict sectional positivity makes the Ricci tensor positive definite, so the Einstein constant is positive. The oriented classification is the principal task. Until the final subsection of Section 8, we therefore work on a connected oriented smooth closed four-manifold, choose an orientation, and rescale the metric so that $\text{Ric} = 3g$. The final orientation-cover argument uses the oriented conclusion proved there and determines the possible nonorientable quotient.

Earlier results. The problem combines rigidity of the Einstein equation with the global restrictions of positive sectional curvature. Berger’s early rigidity work under strict quarter pinching [3] was followed by quantitative sectional-curvature bounds of Yang, Costa, and Cao–Tran [32, 11, 8]. These bounds impose additional quantitative hypotheses, some expressed as upper bounds relative to the Einstein constant rather than as pinching ratios. Gursky–LeBrun prove the Fubini–Study conclusion for compact oriented Einstein manifolds with nonnegative sectional curvature and a nonzero positive-definite intersection form [18, Theorem A]. Koca proves the Fubini–Study conclusion for a complete positively sectionally curved Einstein metric Hermitian for an integrable complex structure [24, Theorem 2]. Under the different condition of nonnegative isotropic curvature, the four-dimensional local-symmetry result is due to Micallef–Wang [26, Theorem 4.4]; Brendle extends the Einstein rigidity result to higher dimensions [6, Theorem 1].

Recent results impose other geometric or topological conditions. Dameno obtains half-conformal flatness from fibrewise constancy of the scalar curvature of an associated twistor metric, and obtains the standard models in the compact positive-scalar case [12, Theorem 1.1 and Corollary 1.2]. Catino–Dameno use separation of the largest and middle eigenvalues of one Weyl block to obtain conformal Kähler geometry after at most a double cover [9, Theorems 1.1 and 1.4]. Di Cerbo treats the closed oriented nonflat case with nonnegative sectional curvature and equality everywhere in a Weyl-norm bound, obtaining a locally symmetric $S^2 \times S^2$ universal cover [14, Proposition 3.2].

Write χ for the Euler characteristic and τ for the signature. Cheng proves the oriented positive-sectional classification under $\chi \leq 3$, and consequently in the presence of a nontrivial Killing field [10, Theorem 1.2 and Corollary 1.4]. His argument combines radial Riccati comparison with Weyl-norm gaps and characteristic-number identities. Gursky–Malchiodi prove the classification for closed oriented Einstein four-manifolds with positive sectional curvature under $2\chi - 3|\tau| \leq 4$, using a Green-function comparison and ADM mass [19, Theorem 1.1]. These cited results retain additional curvature, geometric, topological, or symmetry hypotheses. The argument below addresses the interaction of the two Weyl blocks without such an additional restriction.

The two-block problem. The Hodge star on two-forms has eigenvalues $+1$ and -1 ; its two rank-three eigenbundles are denoted by Λ^+ and Λ^- . In the normalization $\text{Ric} = 3g$, the curvature operator restricts to $\text{Id} + T$ and $\text{Id} + U$ on these bundles, where T, U are symmetric trace-free endomorphisms. They are the two Weyl curvature blocks. The Einstein equation removes the off-diagonal Ricci block but does not force either Weyl block to vanish. Our central intermediate conclusion is *half-conformal flatness*:

$$T \equiv 0 \quad \text{or} \quad U \equiv 0.$$

Use Hilbert–Schmidt norms and put

$$v = |T|/\sqrt{6}, \quad b = |U|/\sqrt{6}, \quad x = -\lambda_{\min}(T), \quad y = -\lambda_{\min}(U).$$

The exact sectional-positivity condition is $x + y < 2$. The inequalities $v \leq x \leq 2v$ and $b \leq y \leq 2b$ imply $v + b < 2$, but this weaker norm condition alone is insufficient for the tensor estimate below. Gursky–LeBrun use the corresponding nonstrict cone and norm inequalities and analyze the case of two nonzero Weyl blocks [18, Lemma 1 and Theorem B]. For a function f , write $\mathbf{E}f$ for its integral divided by total Riemannian volume. The gap and its parallel equality case are established in Gursky–LeBrun [18, Theorem 1]: a nonzero block satisfies $\mathbf{E}v^2 \geq 1$, or the exchanged inequality for b . We derive this statement in our normalization, including zeros of a block.

Scalar coupling of the two blocks also has a clear antecedent. Yang chooses a constant t for which $|W^+| - t|W^-|$ has mean zero and combines the Weyl equation, refined Kato inequality, and Poincaré inequality [32, §6]. Cao–Tran similarly balance regularized powers of the norms [8, Proposition 2.2]. Here a volume and moment argument forces equality of the two second moments, and a weighted Hessian square supplies the additional tensor estimate.

From moment balance to half-conformal flatness. The characteristic identities relate the two second moments to volume and to the dimensions of harmonic self-dual and anti-self-dual two-form spaces. An upper polar-volume deficit, the two gaps, and a strengthened scalar variance inequality force $\mathbf{E}v^2 = \mathbf{E}b^2$ when both blocks are nonzero (Proposition 4.9). This balance controls the mean of $v - b$, enabling the variance inequality to contribute the sign needed in the final integral argument.

The main differential estimate uses the Bianchi identity twice. It first places the gradient of a Weyl block in one irreducible tensor type. The curvature commutator then determines all but

one component of its full, generally nonsymmetric Hessian. A curvature-weighted quadratic form is positive on the remaining component because its lower bound is $4 - 2x - 2y > 0$. Completing its square with different scalar weights for the two blocks and integrating by parts produces inequalities involving only first derivatives (Proposition 5.3). The proof retains both the derivatives of the scalar weights and the derivative of the curvature operator acting on the matrix slot. This local estimate does not use moment balance.

The two weighted inequalities, a scalar identity, and the variance bound produce an integrand $\mathcal{I}$ with nonnegative mean. Explicit rational comparison functions give $c \geq 1/10$, nonnegative quantities r, j built from the norm gradients, and a scalar Θ such that

$$\mathbf{E}\mathcal{I} \geq 0, \quad \mathcal{I} \leq -\frac{(\Theta - cr)^2}{c} - c(j - r^2), \quad r^2 \leq j.$$

The pointwise bounds follow from the weighted Hessian calculation and exact finite polynomial signs. Moreover, $r^2 < j$ almost everywhere that either norm gradient is nonzero. Integration therefore makes both norms constant. If both blocks were nonzero, their separate gaps would then contradict $v + b < 2$. This proves half-conformal flatness.

The final geometric argument has a different role. Once one block vanishes, the characteristic identities leave two harmonic counts. The exceptional count forces $s_0 = \mathbf{E}(v^2 + b^2) = 3$ and volume $4\pi^2/3$; a conditional lower polar-volume estimate excludes precisely that case. Equality in the surviving Weyl gap makes the curvature parallel. Matching its tensor with a standard model and using simple connectivity then gives a global isometry. Figure 1 records the assembly.

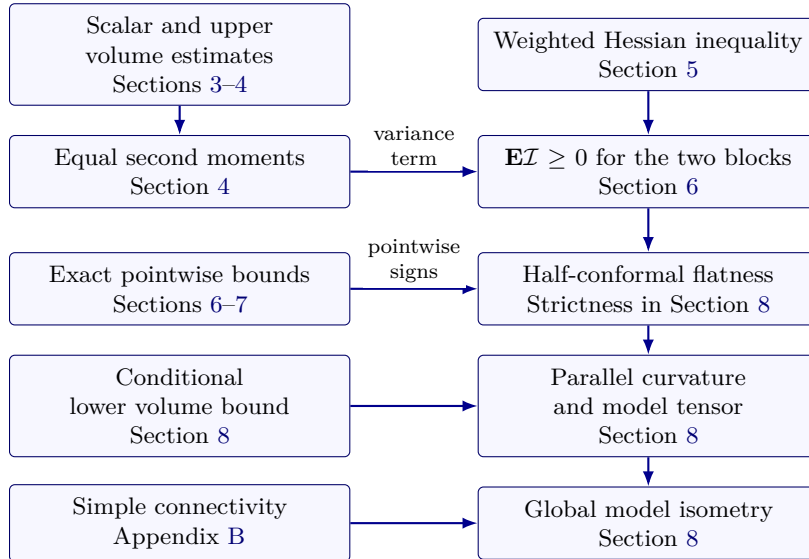


Figure 1: Assembly of the proof. Moment balance enters the nonnegative integral; the exact pointwise bounds and active-gradient strictness force one-block vanishing. The conditional lower bound identifies the remaining parallel tensor, and simple connectivity permits global continuation.

The radial estimates use discrete Jacobi determinants and retain the order of curvature matrices along each geodesic. The lower refinement is conditional on the moment value just described. The article gives the geometric and analytic arguments, together with the finite polynomial formulas and sign proofs. The accompanying checker reproduces the saved finite arithmetic within the precise scope stated in its coverage record.

Organization. Section 2 fixes curvature conventions and derives the differential identities, including their behavior at zeros. Section 3 proves the weighted variance and Weyl-gap estimates and states the characteristic identities. Section 4 proves the upper deficit and moment balance. Section 5 derives the weighted Hessian inequality, and Section 6 combines it with the scalar estimates. Section 7 proves the resulting finite signs. Section 8 proves one-block vanishing, the conditional lower volume bound, and the full metric classification. Appendix A contains the deferred analytic and heat proofs, while Appendix B proves the required topology.

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2 Curvature blocks and their covariant derivatives

Our first task is to express sectional positivity and the differential Bianchi identity in a common set of variables. The curvature decomposition is classical; see Derdziński [13, Section 2]. We give the algebra and fix its normalizations, since the later Hessian estimate depends on the precise representation constants. Rescale the metric so that $\text{Ric} = 3g$, and write

$$d\nu = \frac{d \text{vol}_g}{\text{Vol}(M)}, \quad \mathbf{E}f = \int_M f d\nu.$$

The Laplacian Δ is the trace of the second covariant derivative. Our curvature convention is

$$[\nabla_i, \nabla_j]\xi_k = R_{ijka}\xi_a$$

for a covector, with $R_{ijij} > 0$ on positively curved orthonormal two-planes. All tensor connections are induced by the Levi-Civita connection. We take covariant derivatives before choosing any pointwise eigenbasis.

2.1 The two Weyl blocks

The Hodge-star eigenspaces Λ^+ and Λ^- have eigenvalues $+1$ and -1 , respectively. On an oriented Euclidean four-space, choose triples of skew orthogonal complex structures I_1, I_2, I_3 and J_1, J_2, J_3 satisfying

$$I_1 I_2 = I_3 = -I_2 I_1, \quad J_1 J_2 = J_3 = -J_2 J_1,$$

and their cyclic analogues, with every I_s commuting with every J_t . The two triples can be constructed from left and right quaternionic multiplication. As two-forms, $I_s/\sqrt{2}$ and $J_s/\sqrt{2}$ are orthonormal bases of Λ^+ and Λ^- , respectively. Vector and covector indices are identified by the metric.

Lemma 2.1 (Curvature blocks and sectional positivity). *There are symmetric trace-free endomorphisms $T \in \text{Sym}_0(\Lambda^+)$ and $U \in \text{Sym}_0(\Lambda^-)$ such that the curvature operator is*

$$\mathcal{R}_{\Lambda^2} = \begin{pmatrix} \text{Id} + T & 0 \\ 0 & \text{Id} + U \end{pmatrix}.$$

At a point where these endomorphisms are diagonal, with eigenvalues t_s, u_s , the curvature tensor is

$$R_{ikjl} = \frac{1}{2} \sum_s (1 + t_s) I_{s,ik} I_{s,jl} + \frac{1}{2} \sum_s (1 + u_s) J_{s,ik} J_{s,jl}.$$

Put

$$x = -\min_s t_s, \quad y = -\min_s u_s, \quad v = \frac{|T|}{\sqrt{6}}, \quad b = \frac{|U|}{\sqrt{6}},$$

where the matrix norms are Hilbert–Schmidt norms. Then

$$\sec_g > 0 \iff x + y < 2, \quad v \leq x \leq 2v, \quad b \leq y \leq 2b.$$

Both Weyl blocks, viewed as curvature four-tensors, have zero divergence. In particular, for $Z = \nabla T$, whose matrix entries are one-forms, the differential Bianchi constraint is

$$\sum_s I_s Z_{sr} = 0 \quad (r = 1, 2, 3).$$

Proof. The first Bianchi identity says that the traces of the two diagonal blocks of the curvature operator agree. Their sum is half the scalar curvature, hence is 6. Contraction identifies the off-diagonal block with the trace-free Ricci tensor: its nine entries give the coefficients in the independent symmetric trace-free matrices $I_s J_t$. Thus the Einstein condition makes this block zero, and each diagonal block has trace 3. Expanding in the orthonormal two-form bases gives the displayed curvature tensor.

A unit simple two-form has self-dual and anti-self-dual parts of norm $1/\sqrt{2}$. Conversely, a unit two-form whose two parts have these norms has exterior square zero, and is simple: in its orthogonal normal form $\alpha = a e^1 \wedge e^2 + b e^3 \wedge e^4$, the condition $\alpha \wedge \alpha = 0$ is $ab = 0$. Therefore all pairs of plus and minus parts of norm $1/\sqrt{2}$ occur. The smallest sectional curvature is $1 - (x + y)/2$, proving the equivalence.

For the norm comparison, write the eigenvalues of T as $-x, a, x - a$. Minimality of $-x$ gives $-x \leq a \leq 2x$, and hence

$$\frac{3}{2}x^2 \leq x^2 + a^2 + (x - a)^2 \leq 6x^2.$$

Division by 6 gives $v \leq x \leq 2v$. The proof for U is identical.

Contracting the differential Bianchi identity shows that the curvature tensor is divergence-free when Ricci is parallel. Project its last pair of indices onto Λ^+ or Λ^- ; these projections are parallel. Subtracting the parallel scalar-curvature contribution proves divergence freedom of each Weyl block. In a quaternionic basis, the resulting constraint on ∇T is exactly the stated identity. $\square$

2.2 Algebra of the derivative representations

Let Σ_s denote the infinitesimal action of I_s on the plus indices of a tensor representation. On Λ^+ it is rotation about axis s with speed 2. A complexified irreducible representation has *spin* j when

$$\sum_{s=1}^3 \Sigma_s^2 = -4j(j+1) \text{Id}.$$

Pairs (j, ℓ) specify the spins of the two commuting rotation algebras. This notation describes associated tensor bundles of the oriented orthonormal frame bundle; it requires no spin structure.

Lemma 2.2 (Representation rules). *A spin- j representation has weights $-j, -j+1, \dots, j$ for $\Sigma_3/(2i)$, each once, and*

$$H_j \otimes H_\ell \cong H_{j+\ell} \oplus H_{j+\ell-1} \oplus \dots \oplus H_{|j-\ell|}.$$

In particular, the tangent, plus two-form, and plus trace-free symmetric-matrix representations have types

$$\left(\frac{1}{2}, \frac{1}{2}\right), \quad (1, 0), \quad (2, 0).$$

The Casimir operators and the orthogonal projections onto these irreducible types are parallel bundle endomorphisms.

Proof. The generators are skew-Hermitian and satisfy $[\Sigma_1, \Sigma_2] = 2\Sigma_3$, cyclically. Diagonalize $\Sigma_3/(2i)$ and take the usual conjugate raising and lowering operators, which shift its weight by one. If a unit highest vector has weight j , commuting the raising and lowering operators gives the squared lowering coefficient at weight $j - r$ as

$$(r+1)(2j-r).$$

Nonnegativity and termination of the chain imply that $2j$ is a nonnegative integer and give the listed weights and Casimir. Taking invariant orthogonal complements decomposes an arbitrary representation. Multiplying the weight lists, and successively removing the chain of highest remaining weight, proves the tensor product rule with multiplicity one. Commuting actions operate on the corresponding multiplicity spaces. The same chain description shows that an intertwiner between single irreducible copies is a scalar.

The actions on the tangent space and on Λ^+ give the first two types. The symmetric square of the spin-one representation is spin two plus the scalar trace, giving the third. Finally, these splittings are defined by invariant algebraic operators. The connection induced by the orthonormal frame bundle preserves every such operator. Thus the projections are parallel even though a chosen local frame, or a curvature eigenbasis, need not be parallel. $\square$

Write $F = \text{Sym}_0(\Lambda^+)$. For a symmetric trace-free matrix A , define the self-adjoint operator on F

$$B_F(A)S = \frac{AS + SA}{2} - \frac{\text{tr}(AS)}{3} \text{Id}.$$

It is characterized by $\langle B_F(A)S, S \rangle = \text{tr}(AS^2)$. On any plus spin- j representation with $j \geq 1$, define

$$Q_j(A) = \frac{\sum_{r,s} A_{rs} \Sigma_r \Sigma_s}{4j(2j-1)}.$$

For diagonal A the numerator is $\sum_s a_s \Sigma_s^2$.

Lemma 2.3 (Curvature-weighted operators). *If A has smallest eigenvalue $-x$, then*

$$Q_j(A) \leq \frac{x(j+1)}{2j-1} \text{Id}.$$

Moreover, $Q_2(A) = B_F(A)$. The compression of $B_F(A)$, acting on an F factor, to a highest-spin coupling of total plus spin j is $Q_j(A)$, provided the operator acts trivially on any additional irreducible minus factor.

Proof. In a diagonal basis, $a_s + x \geq 0$ and every Σ_s^2 is nonpositive. Thus

$$\sum_s a_s \Sigma_s^2 \leq -x \sum_s \Sigma_s^2 = 4xj(j+1) \text{Id},$$

which proves the bound.

To identify Q_2 , diagonalize $S = \text{diag}(d_1, d_2, d_3)$. The speed-two matrix action gives $|\Sigma_s S|^2 = 8(d_j - d_k)^2$, with s, j, k cyclic; these three derivatives are mutually orthogonal. Consequently, allowing A to be nondiagonal in this basis,

$$\begin{aligned} \langle Q_2(A)S, S \rangle &= -\frac{1}{3} \sum_s A_{ss} (d_j - d_k)^2 \\ &= -\frac{1}{3} \sum_s A_{ss} (2 \text{tr}(S^2) - 3d_s^2) = \text{tr}(AS^2). \end{aligned}$$

Here trace-freeness of both matrices gives the middle identity and removes the scalar term. Polarization proves the operator equality.

By Lemma 2.2, spin two occurs once in $\text{End}(H_j) \cong H_j \otimes H_j$. Hence an equivariant operator family linear in A is unique up to scalar. The same is true with an irreducible minus factor when

the operators commute with its action. Test the scalar at $A = \text{diag}(-1, -1, 2)$ on a highest vector. At weight m ,

$$Q_j(A) = \frac{j(j+1) - 3m^2}{j(2j-1)},$$

so its highest-weight eigenvalue is -1 . The F highest vector also has $B_F(A)$ eigenvalue -1 . A product of highest vectors is the highest vector of the stretched coupling, proving the claimed compression. For the same reason, compression of a single generator from a spin- ℓ factor to a stretched spin- j product is ℓ/j times the total generator. $\square$

Proposition 2.4 (Differential equations for the Weyl blocks). *The tensor $Z = \nabla T$ belongs to the parallel subbundle G of type $(5/2, 1/2)$. On this bundle, put $B_{\text{op}} = Q_{5/2}(T)$. Then*

$$\Delta T = 6(T - N), \quad N = (T^2)_0 = T^2 - 2v^2 \text{Id},$$

and

$$\Delta Z = (13 - 20B_{\text{op}})Z. \tag{1}$$

Every statement has an exchanged counterpart for U .

Proof. The derivative space $T^*M \otimes F$ has types $(5/2, 1/2)$ and $(3/2, 1/2)$. The divergence takes values in a space whose plus spins are at most $3/2$. It is nonzero: a nonzero covector tensored with $\text{diag}(1, -1, 0)$ has nonzero divergence. It therefore vanishes on the first summand and is injective on the second, by irreducibility. Lemma 2.1 puts Z in the first summand.

Let

$$C = \sum_s I_s \otimes \Sigma_s$$

on $T^*M \otimes F$. Expansion of the combined Casimir gives $C = -4$ on G and $C = 6$ on its complement. In particular $P_G = (6 - C)/10$ is a parallel orthogonal projector. The bundle curvature on F is

$$F_{ik} = \frac{1}{2} \sum_s (1 + t_s) I_{s,ik} \Sigma_s$$

at a point of diagonalization. Taking the covariant divergence of $CZ = -4Z$, using skewness of I_s to take a commutator and $\sum_{ik} I_{s,ik} I_{r,ik} = 4\delta_{sr}$, gives

$$-4\Delta T = \sum_s (1 + t_s) \Sigma_s^2 T = -24T + 24B_F(T)T = -24(T - N).$$

To differentiate this equation, note first that $\nabla N = 2B_F(T)Z$. For each derivative index j , the covariant commutation formula reads

$$\Delta Z_j = \nabla_j \Delta T + \nabla^i (F_{ij} T) + F_{ij} \nabla^i T + \text{Ric}_{ja} Z_a.$$

The divergence of F vanishes because it is the induced action of the divergence-free Riemann curvature. Thus the differentiated equation contributes $6Z - 12B_F(T)Z$, the Ricci term contributes $3Z$, and the remaining bundle term is $-\sum_s (1 + t_s) I_s \Sigma_s Z$. Its constant part is $-CZ = 4Z$.

The left side already lies in G , since P_G is parallel and $P_G Z = Z$. We may therefore compress the right side to G , although its individual summands need not preserve G . Lemma 2.3 gives $P_G B_F(T) P_G = Q_{5/2}(T)$. On G , the total plus generators are $I_s + \Sigma_s$, so

$$40Q_{5/2}(T) = P_G \sum_s t_s (I_s + \Sigma_s)^2 P_G.$$

The I_s^2 term vanishes by $\sum_s t_s = 0$, and the Σ_s^2 term compresses to $24Q_{5/2}(T)$. Hence

$$P_G \sum_s t_s I_s \Sigma_s P_G = 8Q_{5/2}(T).$$

The scalar contributions sum to 13, while the two curvature contributions sum to $-20Q_{5/2}(T)$, proving (1). For a nondiagonal T the same calculation uses the symmetric contraction $\sum_{r,s} T_{rs}(I_r + \Sigma_r)(I_s + \Sigma_s)$: its $I_r I_s$ part is zero by symmetry and trace-freeness. Thus the computation is tensorial and entails no differentiation of a moving eigenframe. $\square$

2.3 The gradient parametrization and the norm equation

Write

$$p = \nabla v, \quad q = \nabla b, \quad e = \frac{|\nabla T|^2}{6}, \quad \bar{e} = \frac{|\nabla U|^2}{6}.$$

The norms v, b are Lipschitz. Their gradients are understood almost everywhere and vanish almost everywhere on the respective zero sets.

Where $v > 0$, diagonalize T and put, cyclically in i, j, k ,

$$t_i = v s_i, \quad \tilde{s}_i = \frac{s_j - s_k}{\sqrt{3}}, \quad \sigma = \frac{1}{6} \sum_i s_i^3, \quad \rho = \frac{1}{6} \sum_i s_i^2 \tilde{s}_i.$$

Here $\sum s_i = 0$ and $\sum s_i^2 = 6$. On the plane of triples with zero sum, the map $s \mapsto \tilde{s}$ is a skew-adjoint orthogonal quarter-turn. Thus

$$\sum_i \tilde{s}_i = 0, \quad \sum_i \tilde{s}_i^2 = 6, \quad \sum_i s_i \tilde{s}_i = 0, \quad s_i^2 + \tilde{s}_i^2 = 4.$$

For later use, these identities imply

$$\sigma^2 + \rho^2 = 1, \quad \frac{N}{v^2} = \text{diag}(\sigma s_i + \rho \tilde{s}_i), \quad \frac{1}{6} \sum_i \tilde{s}_i^3 = -\rho.$$

Indeed, the trace-free triple $n_i = s_i^2 - 2$ has squared norm 6: use $\sum_i s_i^4 = (\sum_i s_i^2)^2/2 = 18$. Its coefficients in the orthogonal basis $s, \tilde{s}$ are σ, ρ , proving the first two assertions. The last follows by factoring the alternating cubic $\sum_i s_i^2(s_j - s_k)$. If $r = x/v$, the other two eigenvalues besides $-r$ have sum r and product $r^2 - 3$, so

$$1 \leq r \leq 2, \quad \sigma = \frac{3r - r^3}{2}.$$

Normalized quantities are used only where the corresponding block is nonzero. Polynomial products such as $\sigma v^3 = \text{tr}(T^3)/6$ have their natural extensions across its zero set. Bars denote the analogous quantities for U .

The next formula gives, in particular, the refined Kato inequality $|\nabla T|^2 \geq (5/3)|\nabla|T||^2$ for a half-Weyl block. Gursky–LeBrun derive this estimate from the same irreducible Bianchi gradient constraint [18, Lemma 4, equations (10)–(12)]; Calderbank–Gauduchon–Herzlich give its general representation-theoretic setting [7, equation (7.1)]. We retain all three vector parameters: the excess over the norm-gradient energy will be needed in the coupled estimate.

Lemma 2.5 (Parametrization of the Bianchi gradient space). *There are unique vectors k, l such that*

$$\begin{aligned} Z_{ii} &= s_i p - \sqrt{3} \tilde{s}_i k, & Z_{jk} &= -I_i \left(l + s_i k + \frac{\tilde{s}_i}{\sqrt{3}} p \right), & (i, j, k) \text{ cyclic}, \\ e &= \frac{5}{3} |p|^2 + 5 |k|^2 + |l|^2. \end{aligned} \quad (2)$$

More generally, for any fixed normalized trace-free diagonal triple s , the same formulas parametrize the entire algebraic space G by three arbitrary real four-vectors.

Proof. Write $D_i = Z_{ii}$ and $a_i = I_i Z_{jk}$. Quaternion multiplication turns the three divergence equations into $D_i + a_j - a_k = 0$. Trace-freeness and the orthogonal basis $s, \tilde{s}$ give uniquely

$$p = \frac{1}{6} \sum_i s_i D_i, \quad k = -\frac{1}{6\sqrt{3}} \sum_i \tilde{s}_i D_i, \quad D_i = s_i p - \sqrt{3} \tilde{s}_i k.$$

Since $s_j - s_k = \sqrt{3} \tilde{s}_i$ and $\tilde{s}_j - \tilde{s}_k = -\sqrt{3} s_i$, the vectors $s_i k + \tilde{s}_i p / \sqrt{3}$ have exactly the differences required of the a_i . Their difference from a_i is therefore a common vector l , proving the formulas and their converse. This also establishes their full algebraic scope.

Orthogonality of $s, \tilde{s}$ gives

$$\sum_i |Z_{ii}|^2 = 6|p|^2 + 18|k|^2, \quad 2 \sum_{j < k} |Z_{jk}|^2 = 4|p|^2 + 12|k|^2 + 6|l|^2.$$

The factor two in the second sum is the Hilbert–Schmidt multiplicity of off-diagonal entries. Division by six proves the norm identity. For $Z = \nabla T$, $\nabla |T|^2 = 2v \sum_i s_i Z_{ii} = 12vp$; comparison with $|T|^2 = 6v^2$ identifies the parameter p with ∇v . $\square$

The pointwise norm equation alone is insufficient for integration: a Weyl block may vanish. We next prove that its regularized Laplacians converge to an actual integrable density, so no measure supported on the zero set is lost.

Proposition 2.6 (Weak norm equation, including the zero set). *The function e/v , defined to be zero on $\{v = 0\}$, belongs to $L^1(M)$, and*

$$\Delta v = 6v(1 - \sigma v) + \frac{e - |p|^2}{v} \quad (3)$$

holds as an equality of distributions with an L^1 density. The quotient is zero almost everywhere on the zero set. This identity may be tested against bounded Lipschitz functions. The corresponding assertions hold for b .

Proof. On $\{v > 0\}$, pair the equation for ΔT with T to obtain

$$\Delta(v^2) = 12(v^2 - \sigma v^3) + 2e.$$

This is globally an identity of smooth functions. Put $v_\varepsilon = \sqrt{v^2 + \varepsilon^2}$. Direct differentiation gives

$$\Delta v_\varepsilon = \frac{6(v^2 - \sigma v^3)}{v_\varepsilon} + q_\varepsilon, \quad q_\varepsilon = \frac{e}{v_\varepsilon} - \frac{v^2 |p|^2}{v_\varepsilon^3}.$$

The refined derivative estimate in (2) yields

$$\frac{2}{5} \frac{e}{v_\varepsilon} \leq q_\varepsilon \leq \frac{e}{v_\varepsilon}.$$

These assertions hold almost everywhere, including the zero set: every smooth component of T has derivative zero almost everywhere on its zero set, hence $Z = 0$ almost everywhere on $\{T = 0\}$. One can see this elementary fact at density points of a level set; a nonzero derivative would instead make that scalar level set a local hypersurface.

The first term in Δv_ε is uniformly bounded in absolute value by $6(v + v^2)$. Its integral is the negative of the integral of q_ε . Consequently $\mathbf{E}(e/v_\varepsilon)$ is uniformly bounded. Monotone convergence proves the asserted L^1 bound for e/v . Off the zero set, q_ε is dominated by e/v and tends to $(e - |p|^2)/v$; on the zero set both are zero almost everywhere. Dominated convergence therefore gives an L^1 limit, rather than only a weak limit of measures. The zeroth-order term converges in L^1 as well. Since $v_\varepsilon \rightarrow v$ uniformly, their Laplacians converge distributionally, proving (3) without a residual singular measure.

Finally, approximate a bounded Lipschitz test function uniformly and in H^1 by smooth functions, using local mollification and a partition of unity. Uniform convergence controls its pairing with the L^1 right side, and H^1 convergence controls its gradient pairing with ∇v . This proves the asserted testing rule. $\square$

3 Scalar estimates and the characteristic identities

This section supplies two kinds of global information: analytic bounds for the sizes of the Weyl blocks, and integer identities relating their second moments to volume. The former will control derivatives in the coupled inequality; the latter will restrict the possible imbalance between the two blocks.

The standing setting is the closed, connected, oriented Einstein four-manifold with $\text{Ric} = 3g$ fixed in the introduction. Appendix A supplies the analytic foundations used here; its proofs depend only on this setting and the curvature conventions of Section 2. We retain the weighted variance argument and the complete Weyl-gap proof in the main text because their precise weights and behavior at zeros are used in the two-block argument.

3.1 A strengthened scalar variance estimate

Proposition 3.1 (Weighted variance). *For every real $a \in H^1(M)$,*

$$\text{Var}(a) := \mathbf{E}[(a - \mathbf{E}a)^2] \leq \mathbf{E} \frac{|\nabla a|^2}{K}, \quad K = 4 + v^2 + b^2. \quad (4)$$

Proof. First let ϕ be smooth and write $S = \text{Hess}_0 \phi$. The integrated Bochner identity and $\text{Ric} = 3g$ give

$$\mathbf{E}[(\Delta \phi)^2] = \mathbf{E} \left(4|\nabla \phi|^2 + \frac{4}{3}|S|^2 \right).$$

Define operators on covariant two-tensors by

$$(T_D)_{ikjl} = \frac{1}{2} \sum_s t_s I_{s,ik} I_{s,jl}, \quad (U_D)_{ikjl} = \frac{1}{2} \sum_s u_s J_{s,ik} J_{s,jl},$$

where ij is the row index and kl is the column index. These tensorial definitions may be evaluated after pointwise diagonalization. They annihilate metric trace. On the symmetric trace-free tensors, the nine matrices $I_i J_j$ form an orthogonal basis, with T_D and U_D eigenvalues t_i and u_j . For example, T_D acts by $-\frac{1}{2} \sum_s t_s I_s(\cdot) I_s$; the quaternion relations give the eigenvalue t_i on $I_i J_j$. Thus

$$T_D + U_D \leq 2(x + y) \text{Id} < 4 \text{Id}$$

on this space.

The integrated curvature contraction has a useful sign and a precise coefficient. Since T_D annihilates trace, replace S by $\text{Hess}\phi$ in its quadratic form. Integrate the derivative k , use $\nabla_k(T_D)_{ikjl} = 0$, and use skewness in i, k to obtain

$$\begin{aligned}\mathbf{E}\langle T_D S, S \rangle &= -\mathbf{E}(T_D)_{ikjl} \phi_l \nabla_k \nabla_i \phi_j \\ &= \frac{1}{2} \mathbf{E}(T_D)_{ikjl} R_{ikja} \phi_a \phi_l.\end{aligned}$$

In this contraction the minus curvature block is orthogonal to the plus block. Substituting their expansions gives

$$\frac{1}{2} \sum_{ikj} (T_D)_{ikjl} R_{ikja} = \frac{1}{2} \sum_s t_s (1 + t_s) \sum_j I_{s,jl} I_{s,ja} = 3v^2 \delta_{la}.$$

The final equality uses $\sum t_s = 0$ and $\sum t_s^2 = 6v^2$. Consequently

$$\mathbf{E}\langle (T_D + U_D) S, S \rangle = 3\mathbf{E}[(v^2 + b^2)|\nabla\phi|^2].$$

The pointwise upper operator bound, inserted into the positive $|S|^2$ term of the Bochner identity, now yields

$$\mathbf{E}[(\Delta\phi)^2] \geq \mathbf{E}K|\nabla\phi|^2.$$

For smooth a , the closed-manifold Poisson statement in Appendix A.1 gives a unique smooth zero-mean solution of $\Delta\phi = a - \mathbf{E}a$. Integration by parts and Cauchy–Schwarz with the pointwise weight K give

$$\begin{aligned}\text{Var}(a) &= -\mathbf{E}\langle \nabla a, \nabla \phi \rangle \\ &\leq \left(\mathbf{E} \frac{|\nabla a|^2}{K} \right)^{1/2} \left(\mathbf{E}K|\nabla\phi|^2 \right)^{1/2} \\ &\leq \left(\mathbf{E} \frac{|\nabla a|^2}{K} \right)^{1/2} \text{Var}(a)^{1/2}.\end{aligned}$$

Division proves the assertion when the variance is nonzero; the zero case is immediate. Smooth functions are dense in H^1 by the local mollification argument recalled in Appendix A.1. Since K is bounded above and below by positive constants, this approximation also passes the weighted energy to the limit and proves the assertion on H^1 . $\square$

3.2 The sharp scalar inequality and the Weyl gap

The following is the four-dimensional Einstein case of Ilias’s sharp positive-Ricci Sobolev inequality [22, Theorem 3]. Its proof in the present normalization is given in Appendix A.2. The nonlinear Hessian argument belongs to the Bochner approach to elliptic rigidity; compare the systematic treatment in [4].

Proposition 3.2 (Normalized Sobolev inequality). *For every real $f \in H^1(M)$,*

$$(\mathbf{E}|f|^4)^{1/2} \leq \mathbf{E} \left(f^2 + \frac{1}{2} |\nabla f|^2 \right).$$

The next proposition is the Weyl-norm gap established in Gursky–LeBrun [18, Theorem 1]; we give the derivation in our normalization, including the equality case and the possible zeros of the Weyl block.

Proposition 3.3 (Weyl-norm gap). *For either Weyl block,*

$$v \equiv 0 \quad \text{or} \quad \mathbf{E}v^2 \geq 1. \quad (5)$$

If $T \not\equiv 0$ and $\mathbf{E}v^2 = 1$, then $\nabla T = 0$ and $v = \sigma = 1$. The corresponding statements hold for U .

Proof. Put $A_\varepsilon = v^2 + \varepsilon^2$ and $f_\varepsilon = A_\varepsilon^{1/6}$. These are smooth functions. The equation for $\Delta(v^2)$ and (2) give

$$\Delta f_\varepsilon = A_\varepsilon^{-5/6} \left[2(v^2 - \sigma v^3) + \frac{1}{3}e - \frac{5}{9} \frac{v^2}{A_\varepsilon} |p|^2 \right] \geq 2A_\varepsilon^{-5/6} (v^2 - v^3).$$

Indeed, $\sigma \leq 1$, $v^2/A_\varepsilon \leq 1$, and $e \geq 5|p|^2/3$. Multiply by f_ε and integrate:

$$\mathbf{E}|\nabla f_\varepsilon|^2 \leq 2\mathbf{E} \frac{v^3 - v^2}{A_\varepsilon^{2/3}} \leq 2\mathbf{E}v^{5/3}.$$

Their L^2 norms are uniformly bounded as well. Thus f_ε is bounded in H^1 , and its uniform limit $f = v^{1/3}$ belongs to H^1 by weak compactness. This bound is obtained directly from the regularized equation; it requires no positive lower bound on v .

Apply Proposition 3.2 to f_ε and use the preceding energy bound:

$$\begin{aligned} (\mathbf{E}f_\varepsilon^4)^{1/2} &\leq \mathbf{E} \left[A_\varepsilon^{1/3} + \frac{v^3 - v^2}{A_\varepsilon^{2/3}} \right] \\ &= \mathbf{E} \frac{\varepsilon^2 + v^3}{A_\varepsilon^{2/3}}. \end{aligned}$$

The term $\varepsilon^2/A_\varepsilon^{2/3}$ is at most $\varepsilon^{2/3}$, while $v^3/A_\varepsilon^{2/3} \leq v^{5/3}$. Dominated convergence therefore gives

$$(\mathbf{E}f^4)^{1/2} \leq \mathbf{E}(f^2v) \leq (\mathbf{E}f^4)^{1/2} (\mathbf{E}v^2)^{1/2}.$$

When $T \not\equiv 0$, the common factor is positive, proving (5).

If $\mathbf{E}v^2 = 1$, equality holds in Cauchy–Schwarz. Thus $v = \alpha f^2$ almost everywhere for a constant $\alpha > 0$. It follows that v takes only the values 0 and α^3 almost everywhere, and hence everywhere by continuity. Connectedness and $T \not\equiv 0$ force the positive constant value on all of M . The moment condition gives $v = 1$. Equation (3) then becomes $0 = 6(1 - \sigma) + e$. Both terms are nonnegative, so $\sigma = 1$ and $e = 0$. Smoothness gives $\nabla T = 0$ everywhere. $\square$

3.3 The Euler and signature formulas

Proposition 3.4 (Characteristic identities). *Let n_+ and n_- be the finite nonnegative integer dimensions of the spaces of harmonic self-dual and anti-self-dual two-forms. The Euler characteristic and signature of M are, respectively,*

$$\chi = 2 + n_+ + n_-, \quad \tau = n_+ - n_-.$$

Then

$$\begin{aligned} 8\pi^2\chi &= 6\text{Vol}(M)(1 + \mathbf{E}(v^2 + b^2)), \\ 12\pi^2\tau &= 6\text{Vol}(M)\mathbf{E}(v^2 - b^2). \end{aligned} \quad (6)$$

If $U = 0$, then $n_- = 0$; likewise $T = 0$ implies $n_+ = 0$.

These are the Euler and signature formulas in the present curvature normalization. Appendix A.3 identifies the topological integers through Hodge theory and proves the formulas by the form-complex heat-supertrace method of McKean–Singer [25]; compare the general formulations and local coefficients in Gilkey [17, Theorems 1.7.6 and 4.8.16]. That appendix also proves the zero-block kernel vanishing statements, which will be needed after one-block vanishing.

4 Upper volume deficit and balanced Weyl moments

Write $V_M = \text{Vol}(M, g)$ and $s_0 = \mathbf{E}(v^2 + b^2)$. Our goal in this section is to prove that the two second moments agree whenever neither Weyl block vanishes. The characteristic identities turn an upper polar-volume deficit into an integer restriction; a weighted scalar variance estimate then excludes the sole remaining unequal-moment case.

Radial comparison also plays a central role in Cheng's classification under the additional assumption $\chi(M) \leq 3$ [10, Proposition 3.1]. His Riccati argument does not require curvature matrices at different times to commute. We use a discrete Jacobi determinant to retain their order and obtain the following moment-dependent deficit.

Proposition 4.1 (Upper volume deficit). *One has*

$$\frac{V_M}{8\pi^2/3} \leq 1 - \frac{3}{40}s_0. \quad (7)$$

Here is how the estimate will be used. If both blocks are nonzero, their gaps and $v + b < 2$ give $2 \leq s_0 < 4$. Equations (6) and (7) then imply $\chi < 7$ and $|\tau|/\chi < 4/15$. The parity and nonnegativity constraints from $n_\pm$ leave only $(\chi, |\tau|) = (5, 1)$ when $\tau \neq 0$. The last part of this section rules out that case and proves Proposition 4.9. We first establish the upper deficit and the common radial machinery; the conditional lower comparison needed after one-block vanishing is proved in Section 8.

Along a unit-speed geodesic, choose a parallel orthonormal normal frame and let $\mathcal{R}(t)$ be the normal Jacobi curvature matrix. Set

$$A(t) = \mathcal{R}(t) - \text{Id}, \quad \mathcal{J}'' + \mathcal{R}\mathcal{J} = 0, \quad \mathcal{J}(0) = 0, \quad \mathcal{J}'(0) = \text{Id}, \quad D(t) = \det \mathcal{J}(t).$$

Positive sectional curvature and $\text{Ric} = 3g$ give $0 < \mathcal{R} < 3\text{Id}$ and $\text{tr } A = 0$. Let $\mathcal{E}$ denote integration over the unit tangent bundle against normalized Riemannian measure on the base and normalized spherical measure in the fibers.

Lemma 4.2 (Geodesic averages). *At every time along the geodesic flow,*

$$\mathcal{E}|A|^2 = \frac{3}{2}s_0, \quad \mathcal{E}|A'|^2 \leq \frac{3}{2}\mathcal{E}|A|^2. \quad (8)$$

Proof. The indicated probability measure is invariant under geodesic flow. Indeed, in cotangent coordinates the Hamiltonian flow of kinetic energy preserves canonical volume: its divergence vanishes by equality of the mixed second derivatives of the Hamiltonian. Applying this observation to energy shells gives invariance of the measure on the unit-energy level. In orthonormal fiber coordinates this is precisely Riemannian base volume times spherical measure.

The plus and minus contributions to A are orthogonal conjugates of $T/2$ and $U/2$, using the normal bases $I_s\dot{\gamma}$ and $J_s\dot{\gamma}$. The same description applies to their derivatives in parallel frames. The mixed contractions have zero spherical average: they are invariant bilinear pairings between the inequivalent types $(2, 0)$ and $(0, 2)$, or between $(5/2, 1/2)$ and $(1/2, 5/2)$. Consequently,

$$\mathcal{E}|A|^2 = \frac{1}{4}\mathbf{E}(|T|^2 + |U|^2), \quad \mathcal{E}|A'|^2 = \frac{1}{16}\mathbf{E}(|\nabla T|^2 + |\nabla U|^2).$$

Here the second factor $1/4$ in the derivative identity is the spherical average of a squared directional derivative in four dimensions. Integrating the equation for ΔT yields

$$\mathbf{E}e = 6\mathbf{E}(\sigma v^3 - v^2) \leq 6\mathbf{E}v^2,$$

since $\sigma \leq 1$ and $v < 2$. The exchanged identity gives the same bound for $\bar{e}$. The stated formulas follow. $\square$

4.1 Minimizing geodesics and upper polar integration

The global input is the classical index-form and closed-geodesic argument of Myers, Synge, and Klingenberg [27, 31, 23]; see also [1, Theorem 1.5 and Remark 1.2(ii)]. Its full proof is in Appendix B, under all the standing oriented, closed, normalized positivity hypotheses.

Lemma 4.3 (Diameter, injectivity, and topology). *The diameter of M is at most π , its injectivity radius is at least $\pi/\sqrt{3}$, and M is simply connected. Every minimizing geodesic segment has no point conjugate to its initial point in its interior.*

The last assertion allows a conjugate endpoint. It is needed in the upper polar argument even for segment lengths between $\pi/\sqrt{3}$ and π .

Put $D_c(t) = D(t)$ before the first conjugate time and $D_c(t) = 0$ at and after it.

Lemma 4.4 (Upper polar bound). *One has*

$$V_M \leq 2\pi^2 \int_0^\pi \mathcal{E} D_c(t) dt.$$

Proof. Fix a center p and take all vectors of length less than π whose radial geodesics have no conjugate point at or before their endpoints. This is an open set: the Jacobi determinant divided by t^3 , continued as 1 at zero, has a positive minimum on each such radial interval. Every point is the exponential image of a minimizing vector of length at most π , by Lemma 4.3. A minimizing vector is in this open set unless it is critical for $\exp_p$, has length π , or is zero.

The exponential map is smooth with bounded derivative on the closed radius- π ball. The area formula with multiplicity states that the integral of its absolute Jacobian over a measurable set equals the integral of the number of preimages over its image. In particular, critical vectors and the boundary sphere have images of zero four-dimensional volume: their respective Jacobian integral and domain volume are zero. On the open preconjugate set the determinant is positive. The area formula therefore bounds V_M by the integral of D over this set, with multiplicities only increasing that integral. Nonconjugate cut vectors are included in the set; no identification of cut points with conjugate points is used. Polar coordinates give the claimed bound at each center, since the area of the unit three-sphere is $2\pi^2$. Average over centers and use Tonelli's theorem. $\square$

4.2 Discrete determinant comparisons

The upper polar bound reduces the volume problem to estimates for a matrix Jacobi determinant. We approximate the Jacobi equation by a finite Dirichlet matrix. Its Schur complements and sine-mode compressions retain the order of matrix multiplication, without requiring simultaneous diagonalization along the geodesic. The same discrete determinant will be used again for the conditional lower comparison in Section 8.

Lemma 4.5 (The discrete Jacobi determinant). *Fix a smooth symmetric normal curvature matrix on $[0, t]$, put $a = t/N$, and let $\mathcal{L}_N$ be the block tridiagonal matrix on nodes $1, \dots, N-1$ with diagonal blocks $2\text{Id} - a^2\mathcal{R}(ia)$ and off-diagonal blocks $-\text{Id}$. Then*

$$a^3 \det \mathcal{L}_N \longrightarrow D(t).$$

If t is before the first conjugate time, $\mathcal{L}_N$ is positive definite for all sufficiently large N . The Jacobi convergence is uniform over families with uniformly bounded coefficients and first derivatives.

Proof. Define matrices without using inverses by

$$Y_0 = 0, \quad Y_1 = \text{Id}, \quad Y_{i+1} = (2\text{Id} - a^2\mathcal{R}(ia))Y_i - Y_{i-1}.$$

The position aY_i and backward velocity $Y_i - Y_{i-1}$ evolve by first changing velocity by $-a\mathcal{R}(ia)(aY_i)$ and then changing position by a times the new velocity. On the exact Jacobi solution one step has error $O(a^2)$, while the update amplifies errors by at most $1 + Ca$. Iteration proves convergence of position and velocity, uniformly on the fixed interval. The constants are uniform under the stated coefficient bounds.

Suppose first that t is preconjugate. Away from zero the exact Jacobi matrix is invertible with uniformly bounded inverse on compact time intervals, so the converging aY_i are invertible there. Near zero use the identity, still without inverses,

$$Y_i = i\text{Id} - a^2 \sum_{r < i} (i - r)\mathcal{R}(ra)Y_r.$$

For sufficiently small fixed ia it gives $\max_{r \leq i} |Y_r| \leq Ci$ by absorbing the error, and then $Y_i/i = \text{Id} + O((ia)^2)$. Thus all the early matrices are invertible as well. All inverses needed for elimination have now been obtained independently of that elimination.

The Schur blocks are

$$S_i = Y_{i+1}Y_i^{-1} = 2\text{Id} - a^2\mathcal{R}(ia) - S_{i-1}^{-1}.$$

They are symmetric, starting with $S_1 = 2\text{Id} - a^2\mathcal{R}(a)$. Near zero they differ from $((i+1)/i)\text{Id}$ by $O((ia)^2 + a^2)$; away from zero, convergence of position and velocity gives $S_i = \text{Id} + O(a)$. Choose the small initial interval first, and then N large. All Schur blocks are positive definite. Block elimination proves positivity of $\mathcal{L}_N$ and gives

$$\det \mathcal{L}_N = \prod_{i=1}^{N-1} \det S_i = \det Y_N.$$

The desired limit follows. The determinant identity holds for arbitrary coefficients by continuity, since invertible Schur blocks form a dense set; convergence therefore gives the limit without the preconjugacy assumption as well. $\square$

Lemma 4.6 (Upper radial comparison). *For $0 < t < \pi$, set*

$$\beta = \frac{t^2}{\pi^2 - t^2}, \quad B = \frac{2}{t} \int_0^t A(s) \sin^2(\pi s/t) ds.$$

Before conjugacy,

$$D(t) \leq \sin^3 t \det(\text{Id} - \beta B), \quad \text{Id} - \beta B \succeq 0.$$

At every time $0 < t < \pi$,

$$1 - \frac{D_c(t)}{\sin^3 t} \geq \frac{1}{6} \min\{1, \beta^2 \max(1, 3 - \beta)\} |B|^2.$$

Proof. Compare $\mathcal{L}_N$ with the matrix for $\mathcal{R} = \text{Id}$, whose scalar sine-mode eigenvalues are $2 - 2\cos(k\pi/N) - a^2$. They are positive for large N because $t < \pi$. Normalize $\mathcal{L}_N$ by the inverse square root of this comparison matrix. The resulting positive matrix has trace 3 on every diagonal sine-mode block, since $\text{tr } A(ia) = 0$.

Split off its first three-dimensional mode and write the normalized matrix as $\begin{pmatrix} P & C \\ C^* & Q \end{pmatrix}$. Its Schur complement is positive and bounded above by Q . Thus

$$\det \begin{pmatrix} P & C \\ C^* & Q \end{pmatrix} = \det P \det(Q - C^* P^{-1} C) \leq \det P \det Q \leq \det P.$$

The last inequality is arithmetic–geometric mean, since $\text{tr } Q = \dim Q$. The first-mode compression tends to $\text{Id} - \beta B$, because its normalized sine vector has entries $\sqrt{2/N} \sin(\pi i/N)$ and

$$\frac{a^2}{2 - 2 \cos(\pi/N) - a^2} \longrightarrow \beta.$$

Lemma 4.5 gives the determinant ratio limit $D(t)/\sin^3 t$. Positive compressions have a positive semidefinite limit, and no inverse of that limiting compression is used. This proves the first comparison.

The eigenvalues of B lie in $[-1, 2]$ and sum to zero, so $|B|^2 \leq 6$. Before conjugacy the determinant loss is

$$1 - \det(\text{Id} - \beta B) = \frac{\beta^2}{2} |B|^2 + \beta^3 \det B.$$

Only $\det B < 0$ is unfavorable. Write the eigenvalues then as $-r, a_1, a_2$, where $a_1, a_2 \geq 0$ and $a_1 + a_2 = r$. One has

$$|\det B| \leq \frac{r}{6} |B|^2, \quad r \leq 1, \quad \beta r \leq 2.$$

The first inequality follows from $(a_1 - a_2)^2 \geq 0$; the last follows from $\beta a_i \leq 1$. These inequalities give the displayed loss bound. If $\det B \geq 0$ the same bound is immediate. After conjugacy its left side is 1, so $|B|^2 \leq 6$ gives it directly. $\square$

4.3 An exact constant for the upper deficit

We use one rational sine minorant for both the upper calculation here and the conditional lower calculation in Section 8.

Lemma 4.7 (A rational sine minorant). *Put*

$$a_0 = \frac{157}{50}, \quad b_0 = \frac{22}{7}, \quad S(z) = a_0 z - \frac{(b_0 z)^3}{6} + \frac{(a_0 z)^5}{120} - \frac{(b_0 z)^7}{5040}.$$

The constants satisfy $a_0 < \pi < b_0$, and $0 \leq S(z) \leq \sin(\pi z)$ for $0 \leq z \leq 1/2$. Its coefficients are

$$S(z) = \frac{157}{50} z - \frac{5324}{1029} z^3 + \frac{95388992557}{37500000000} z^5 - \frac{155897368}{259416045} z^7.$$

Proof. The elementary bounds $a_0 < \pi < b_0$ and the alternating sine expansion through the negative seventh-degree term give $S(z) \leq \sin(\pi z)$ on this interval. Also

$$S(z) \geq z \left(a_0 - \frac{b_0^3}{24} - \frac{b_0^7}{322560} \right) \geq \frac{9}{5} z \geq 0.$$

Expanding the four powers gives the displayed rational coefficients. $\square$

Define

$$f(u) = \begin{cases} u^4(3 + 5u^2 + 6u^4), & 0 \leq u \leq 5/8, \\ 15/16, & 5/8 < u \leq 1. \end{cases}$$

The following strict rational lower bounds include the prefactor. The rows use the indicated polynomial representatives up to their shared endpoints, which do not affect the integrals.

Integrand and prefactor	Interval	Degree	Lower bound
$\frac{9a_0}{48} f(u)(1 - u^2/2)S(\min(u, 1 - u))^3$	$[0, 1/2]$	31	$11869/10^6$
same	$[1/2, 5/8]$	31	$30175/10^6$
same	$[5/8, 1]$	23	$39518/10^6$

For a rational polynomial $p(u) = \sum c_j u^j$ on $[r, s]$, these values are obtained from $\int_r^s p(u) du = \sum c_j (s^{j+1} - r^{j+1})/(j+1)$. The three rows sum to more than $163/2000 > 2/25$; the cubic power S^3 is essential in each row.

Proof of Proposition 4.1. For the sine-squared probability weight defining B , its time variance is $t^2(1/12 - 1/(2\pi^2))$. The variance identity and (8) give

$$\mathcal{E}|B|^2 \geq \mathcal{E}|A|^2 \left(1 - \frac{3}{2}t^2 \left(\frac{1}{12} - \frac{1}{2\pi^2} \right) \right).$$

Indeed, parallel identification along a geodesic and Cauchy-Schwarz show $\mathcal{E}|A(s) - A(r)|^2 \leq (s - r)^2 \mathcal{E}|A'|^2$. With $u = t/\pi$ the factor in parentheses is at least $1 - u^2/2$. For $0 \leq u < 1$, furthermore,

$$f(u) \leq \min\{1, \beta^2 \max(1, 3 - \beta)\}, \quad \beta = \frac{u^2}{1 - u^2}.$$

At $u = 1$ we give the capped comparison function its continuous value 1. For $u \leq 5/8$, comparison with $\beta^2(3 - \beta)$ reduces to $6 - 13u^2 + 6u^4 \geq 0$. On this interval f is increasing and $f(5/8) = 7511875/8388608 < 1$, which also verifies the cap by 1. The comparison function $\min\{1, \beta^2 \max(1, 3 - \beta)\}$, as a function of u , is nondecreasing and remains 1 after it first reaches 1; at $u = 5/8$ its value is $57500/59319 > 15/16$, which handles the rest of the interval. Lemmas 4.4 and 4.6 therefore show that the coefficient of s_0 saved from the round volume is at least

$$\eta_+ = \frac{9\pi}{48} \int_0^1 f(u)(1 - u^2/2) \sin^3(\pi u) du.$$

The first three rational bounds above give $\eta_+ > 2/25 > 3/40$, proving (7). □

4.4 A moment obstruction

The upper volume deficit is proved. Together with the characteristic identities and the two norm gaps, it leaves a single possible unequal-moment integer case when both blocks are nonzero, up to exchanging the blocks. The next scalar lemma will exclude that case. Its proof reduces a constrained probability measure to two support points; all numerical signs are established by explicit rational coefficients.

For $0 \leq z \leq 2$, define

$$H(z) = \sqrt{z} \left(1 + \frac{z^2}{40} - \frac{z^4}{1152} \right), \quad \Phi(z) = 16H(z)^2 + 9z(1 - z).$$

Lemma 4.8 (Two constrained moments). *Every probability measure μ on $[0, 2]$ satisfying*

$$\int z^2 d\mu \geq 1, \quad \int (28z + 6z^2) d\mu \leq 25$$

satisfies

$$16 \left(\int H d\mu \right)^2 - \int \Phi d\mu < 0.$$

Proof. The feasible measures form a nonempty weakly compact convex set. Suppose the continuous functional $J(\mu) = 16(\int H d\mu)^2 - \int \Phi d\mu$ has a nonnegative maximum, attained at μ , and set $d = \int H d\mu$. First $d \geq 11/20$. Indeed,

$$1 + z^2/40 - z^4/1152 \geq 1 + (31/1440)z^2,$$

and

$$25 + \frac{31}{45}z^2 - \frac{346}{25}z \geq \frac{17}{225} > 0 \quad (0 \leq z \leq 2).$$

The last quadratic is decreasing on this interval and has the stated value at 2. It follows that $\Phi(z) \geq 16(11/20)^2 z^2$. The moment constraint and $J(\mu) \geq 0$ now give the claim about d .

For another feasible measure ν , direct expansion with $\ell = 32dH - \Phi$ gives

$$J((1-t)\mu + t\nu) - J(\mu) = t \left(\int \ell d\nu - \int \ell d\mu \right) + 16t^2 \left(\int H d\nu - d \right)^2.$$

For $t \downarrow 0$, maximality shows that μ maximizes the linear functional $\int \ell$. This conclusion does not require concavity of J .

Put $g_1(z) = 28z + 6z^2 - 25$ and $g_2(z) = 1 - z^2$. The measure with mass $9/32$ at 2 and the rest at 0 has respective constraint expectations $-5/2$ and $-1/8$, so both constraints are strictly feasible. The compact convex set of triples

$$\left\{ \left(\int g_1 d\nu, \int g_2 d\nu, \int \ell d\nu \right) : \nu \text{ a probability measure on } [0, 2] \right\}$$

is disjoint from the open set of triples with first two coordinates negative and third coordinate greater than $\int \ell d\mu$. Separation gives coefficients $-\alpha, -\gamma, \beta$, with $\alpha, \gamma, \beta \geq 0$. The last coefficient cannot be zero: otherwise the strictly feasible triple would violate separation. Normalize $\beta = 1$. Evaluating on μ and on point masses shows that μ is supported on the global maxima of

$$F_d(z) = 32dH(z) - \Phi(z) - \alpha(28z + 6z^2) - \gamma(1 - z^2), \quad \alpha, \gamma \geq 0.$$

For completeness, separation bounds $\int (\ell - \alpha g_1 - \gamma g_2) d\nu$ by $\int \ell d\mu$ for every ν ; for μ the reverse inequality follows from its nonpositive constraints. Equality and then point masses give precisely the asserted support condition.

We next verify that $F_d'''(z) > 0$ on $(0, 2]$. Set

$$L(z) = \frac{3}{8} + \frac{3z^2}{64} - \frac{35z^4}{1024}, \quad V(z) = \frac{24}{5} - \frac{16z^2}{15} - \frac{7z^4}{48} + \frac{7z^6}{1152}.$$

Direct differentiation gives $H'''(z) = L(z)z^{-5/2}$ and $\Phi'''(z) = V(z)$. For $x = \sqrt{z/2}$, the polynomial inequalities

$$L(2x^2) > 0, \quad D_*(x) := \frac{88}{5}L(2x^2) - \frac{17}{3}x^5V(2x^2) > 0 \quad (0 \leq x \leq 1)$$

are certified below. Since $32d \geq 88/5$ and $4\sqrt{2} < 17/3$, they imply $32dL(z) - z^{5/2}V(z) > 0$ when $V(z) \geq 0$; when $V(z) < 0$ it is immediate. The constraint polynomials have zero third derivative, proving the claim.

Also $F'_d(0+) = +\infty$. Since F''_d is strictly increasing, F'_d can cross from positive to negative at most once. There is at most one interior local maximum; an interval of maxima is impossible. The endpoint 0 is not a maximum, so the support contains at most one interior point and the endpoint 2. A point mass is infeasible: its first moment condition would give $z \geq 1$, and then $28z + 6z^2 \geq 34$. Thus the support is $\{w, 2\}$, with weight t at 2. Necessarily $0 < w < 1$, and the moment constraints become

$$t_-(w) := \frac{1 - w^2}{4 - w^2} \leq t \leq t_+(w) := \frac{25 - 28w - 6w^2}{80 - 28w - 6w^2}.$$

The denominators are positive. Nonemptiness reduces to $(2-w)(10-37w) \geq 0$, hence $0 < w \leq 10/37$.

For fixed w , J is a convex quadratic in t , so it is bounded above by the larger of its two endpoint values. Those values are negative:

$$16t(1-t)(H(2) - H(w))^2 > 9((1-t)(w^2 - w) + 2t) \quad (t = t_-(w), t_+(w)).$$

To verify these inequalities exactly, put

$$x = \sqrt{w/2} \leq \frac{37}{100}, \quad R_*(x) = 1 + \frac{x^4}{10} - \frac{x^8}{72}.$$

For each of

$$(A_-, B_-) = (1 - w^2, 4 - w^2), \quad (A_+, B_+) = (25 - 28w - 6w^2, 80 - 28w - 6w^2), \quad w = 2x^2,$$

the required inequality, after multiplication by its positive $B_\pm^2$, is $P_\pm(x) > 0$, where

$$P_\pm(x) = 32A_\pm(B_\pm - A_\pm)(R_*(1) - xR_*(x))^2 - 9((B_\pm - A_\pm)(w^2 - w) + 2A_\pm)B_\pm.$$

This uses $H(2) - H(w) = \sqrt{2}(R_*(1) - xR_*(x))$. The endpoint $w = 10/37$, at which the weight interval can collapse, is included in the following strict certificates.

Here are all the coefficient bounds used in this proof. If $p(r + (s - r)y) = \sum_{j=0}^n a_j y^j$, define

$$h_i = \sum_{j=0}^i a_j \frac{\binom{i}{j}}{\binom{n}{j}} \quad (0 \leq i \leq n).$$

Binomial expansion gives $p(r + (s - r)y) = \sum_i h_i \binom{n}{i} y^i (1 - y)^{n-i}$; thus positive h_i give positivity on $[r, s]$. If the original coefficients are $p(x) = \sum p_k x^k$, they are obtained explicitly from

$$a_j = (s - r)^j \sum_{k=j}^n p_k \binom{k}{j} r^{k-j}.$$

The degrees and strict lower bounds for the resulting coefficients are

Polynomial	Degree	Interval	Lower bound
$L(2x^2)$	8	$[0, 1]$	$1/100$
$D_*(x)$	17	$[j/4, (j+1)/4], j = 0, 1, 2, 3$	$2/5$
$P_-(x)$	22	$[0, 37/100]$	1
$P_+(x)$	22	$[0, 37/100]$	1

Equivalently,

$$J := 16(\mathbf{E}H(b))^2 - \mathbf{E}\Phi(b) \geq 0, \quad \Phi(b) = 16H(b)^2 + 9b(1-b). \quad (9)$$

The distribution of b is a probability measure on $[0, 2]$ satisfying the two moment constraints. Lemma 4.8 contradicts (9). Thus $\tau = 0$, and (6) gives equality of the two second moments. $\square$

5 A weighted Hessian inequality for the Weyl blocks

We continue the case in which neither Weyl block vanishes identically. The preceding argument gives $\mathbf{E}v^2 = \mathbf{E}b^2$. This equality will enter the coupling in Section 6; the weighted Hessian inequality proved here does not itself require it. Our task is to extract a first-derivative inequality from the Bianchi-constrained second derivatives. Unequal weights on the two blocks will create a useful interaction, but their derivatives have costs that must be retained exactly. Set $\delta = 2 - v - b$, and retain the notation $Z = \nabla T$, $p = \nabla v$, $q = \nabla b$, and $e = |Z|^2/6$. The argument is stated for T ; exchanging the two orientations gives its companion for U . All expectations use normalized Riemannian volume. The local square will first be proved for an arbitrary global positive Lipschitz weight. We will choose the two fixed weights only when their derivative costs enter.

5.1 The constrained Hessian and its curvature operator

Retain $F = \text{Sym}_0(\Lambda^+)$ and the Bianchi gradient bundle $G \subset T^*M \otimes F$ from Section 2. Their types are $(2, 0)$ and $(5/2, 1/2)$. The irreducible gradient constraint and the refined Kato coefficient used below are established in Gursky–LeBrun [18, Lemma 4, equations (10)–(12)]; Calderbank–Gauduchon–Herzlich [7] place them in the general generalized-gradient framework. We use the parallel projector $P_G = (6 - C)/10$ and the compression identities already proved in Proposition 2.4. They are pointwise tensor identities, including at repeated eigenvalues, and do not require a smoothly chosen eigenbasis.

Lemma 5.1 (The two Hessian components). *Let $\mathcal{H} = T^*M \otimes G$, and write $X = \nabla Z$, so $X_{ij} = \nabla_i \nabla_j T$. Then*

$$X = Y + D_0, \quad Y \text{ has type } (3, 1), \quad D_0 \text{ has type } (2, 0).$$

If $\text{Tr} : \mathcal{H} \rightarrow F$ is contraction of the two derivative slots and Tr^ its adjoint, then*

$$\text{Tr} \text{Tr}^* = \frac{12}{5} \text{Id}, \quad D_0 = \frac{5}{12} \text{Tr}^* \Delta T, \quad |D_0|^2 = \frac{5}{12} |\Delta T|^2.$$

All the type projections in this statement are parallel.

Proof. The product rule gives four inequivalent, multiplicity-one summands

$$\mathcal{H} = (3, 1) \oplus (3, 0) \oplus (2, 1) \oplus (2, 0).$$

Skewing the derivative slots takes values in

$$\Lambda^2 T^*M \otimes F = (3, 0) \oplus (2, 0) \oplus (1, 0) \oplus (2, 1).$$

For the actual Hessian $X = \nabla \nabla T$, its skew difference is $[\nabla_i, \nabla_j]T = F_{ij}T$, where F_{ij} is the curvature of the bundle F . It is essential that the tensor on which this curvature acts is the same Weyl block T that occurs in F_{ij} . The minus rotation algebra acts trivially on F , so there is no minus component. The dependence on T is an equivariant linear term and an equivariant term in its symmetric square.

The latter has plus spins 4, 2, 0; its only common type with $\Lambda^+ \otimes F = (3, 0) \oplus (2, 0) \oplus (1, 0)$ is (2, 0). The linear term has that type as well. Thus the skew commutator has neither a (3, 0) nor a (2, 1) component. This conclusion is specific to the Hessian of T , not to the derivative of an arbitrary section of G .

Skewing is injective on each of those two domain summands. The (3, 0) summand is entirely skew: its stretched plus coupling makes the two derivative plus spin halves symmetric, while their minus coupling to spin zero is alternating. To treat (2, 1), suppose that a tensor of this type is symmetric in its full derivative slots. Its derivative minus pair has spin 1 and is symmetric, so its derivative plus pair must also be symmetric and have spin 1. The inner Bianchi constraint makes the coupling operator $\sum I_s^{(j)} \Sigma_s$ equal to -4 . Symmetry in the derivative slots gives the same value for $\sum I_s^{(i)} \Sigma_s$. The total plus Casimir would therefore be

$$-8 - 24 + 2(-4 - 4) = -48,$$

whereas total spin 2 requires -24 . This contradiction proves injectivity. Moreover, the two images lie in distinct target isotypic components, so they cannot cancel one another. The corresponding components of X must vanish separately.

For the trace calculation, let $K : F \rightarrow T^*M \otimes T^*M \otimes F$ be $K(S)_{ij} = \delta_{ij}S$. The adjoint of the restricted trace is $\text{Tr}^* = P_G^{(j)} K$, where the projector acts on the inner derivative and matrix slots. Now

$$\text{Tr } K = 4\text{Id}, \quad \text{Tr}(C^{(j)}K(S)) = \sum_{i,s} (I_s)_{ii} \Sigma_s S = 0.$$

Thus $\text{Tr } \text{Tr}^* = 12\text{Id}/5$. By equivariance, trace vanishes on (3, 1) and is supported on the unique (2, 0) summand. The displayed nonzero composition makes its restriction to that five-dimensional summand an isomorphism onto F . It therefore determines the whole of D_0 , including its skew part, and gives both the formula and the norm identity. One can also see explicitly that the trace component retains the curvature commutator. At a point, its adjoint is

$$(\text{Tr}^* S)_{ij} = \frac{3}{5} \delta_{ij} S + \frac{1}{10} \sum_s I_{s,ij} \Sigma_s S.$$

Hence

$$(D_0)_{ij} - (D_0)_{ji} = \frac{1}{12} \sum_s I_{s,ij} \Sigma_s \Delta T = \frac{1}{2} \sum_s (1 + t_s) I_{s,ij} \Sigma_s T.$$

For the last equality use $\Delta T = 6(T - N)$ and, in a pointwise diagonal frame, $\Sigma_s N = -t_s \Sigma_s T$. The invariant version is $\Sigma_r N = -\sum_s T_{rs} \Sigma_s T$; no eigenframe has been differentiated. Thus the displayed skew difference is exactly the bundle-curvature action.

Finally, the type projections are invariant expressions in the combined Casimirs, and trace is a metric contraction. They are parallel and do not depend on any curvature eigenframe. The spin labels describe tensor representations of the oriented frame group; no spin structure is used. $\square$

The derivative-slot operators T_D, U_D were defined in the weighted variance proof, and $B_F(T)$ acts on the matrix slot. Compress all three orthogonally to $\mathcal{H} = T^*M \otimes G$. The useful combination is visible before the quadratic form is chosen. Let Σ_s^{tot} be the total plus generator on $\mathcal{H}$ and put

$$\tilde{Q} = \frac{1}{60} \sum_{r,s} T_{rs} \Sigma_r^{\text{tot}} \Sigma_s^{\text{tot}}.$$

Compressing the inner derivative generator to G gives $I_s^{(j)} = \Sigma_s^G/5$: equivariance fixes the spin-one family up to a scalar, and the highest weight gives $(1/2)/(5/2) = 1/5$. At a point where T is diagonal, expansion of the total square therefore gives

$$T_D = \frac{1}{10} \sum_s t_s I_s^{(i)} \Sigma_s^G = 3\tilde{Q} - 2Q_{5/2}(T), \quad B_F(T) = Q_{5/2}(T)$$

on the compressed inner factor. The inner square contributes $40Q_{5/2}(T)$, while the trace-free outer half-spin square is zero. Consequently

$$T_D + 2B_F(T) = 3\tilde{Q}.$$

This combined identity cancels the inner curvature action. On the free $(3, 1)$ type, $\tilde{Q} = Q_3(T) \leq 4x/5$ and $U_D \leq 2y$. Thus the choice

$$\mathcal{P} = 4 - \frac{5}{6}(T_D + 2B_F(T)) - U_D = 4 - \frac{5}{2}\tilde{Q} - U_D$$

has its plus and minus losses bounded by $2x$ and $2y$, respectively. The scalar 4 then meets the sectional boundary $x + y = 2$. This explains the useful coefficients; no uniqueness of the choice is asserted.

Lemma 5.2 (Positivity on the highest Hessian type). *The combined compressed operator $\mathcal{P}$ preserves all four types in $\mathcal{H}$. It is positive definite on $(3, 1)$, and on the trace type it satisfies*

$$\mathcal{P} \operatorname{Tr}^* = \operatorname{Tr}^*(4 - B_F(T)).$$

The individual derivative-slot and matrix-slot summands need not preserve the four types.

Proof. The total plus generators preserve each joint type. Similarly, $\sum_s u_s (J_s^{(i)} + J_s^{(j)})^2 = 4U_D$, so U_D vanishes on minus spin zero and is $Q_1(U)$ on minus spin one. Its latter spectrum is the spectrum of U . The displayed combined formula for $\mathcal{P}$ therefore preserves the joint types. On $(3, 1)$,

$$\mathcal{P} \geq 4 - 2x - 2y > 0.$$

On $(2, 0)$ the denominator in Q_2 is 24, so under the trace identification $\tilde{Q} = (2/5)Q_2(T) = (2/5)B_F(T)$ and $U_D = 0$. This proves the stated trace action. $\square$

5.2 The weighted square and its first-derivative form

The undetermined Hessian component Y is now isolated, and its curvature-weighted quadratic form is positive. We complete its weighted square and integrate by parts to replace the Hessian by first derivatives. The prescribed trace component must be subtracted in full.

Set

$$B = \frac{\langle B_F(T)Z, Z \rangle}{6}, \quad C_N = \frac{\langle B_F(N)Z, Z \rangle}{6}, \quad E_B = \frac{|Q_{5/2}(T)Z|^2}{6}, \quad N = (T^2)_0.$$

For a scalar weight m , write $L_m = dm \otimes Z$, with no symmetrization, and let $(L_m)_Y$ denote its $(3, 1)$ projection.

Proposition 5.3 (Weighted Hessian inequality). *For every global Lipschitz scalar weight $m \geq m_0 > 0$,*

$$0 \leq \mathbf{E} \left[m G_v(p, k, l) - \frac{5}{6} \sum_i m_i \frac{\text{tr}(Z_i \sum_j Z_j^2)}{6} + \frac{\langle \mathcal{P}(L_m)_Y, (L_m)_Y \rangle}{24m} \right], \quad (10)$$

where i, j in the middle term are derivative indices and

$$G_v = \left(-42 - 3b^2 - \frac{10}{3}v^2 \right) e + 75B + \frac{25}{3}C_N - \frac{100}{3}E_B + 5v^2|p|^2.$$

Proof. By Lemma 5.2,

$$\mathbf{E} \left[\frac{m}{6} \left\langle \mathcal{P} \left(Y + \frac{(L_m)_Y}{2m} \right), Y + \frac{(L_m)_Y}{2m} \right\rangle \right] \geq 0.$$

The last term in its expansion is the last term of (10). Since $\mathcal{P}$ preserves the type splitting and $X = Y + D_0$, the other terms are

$$\frac{1}{6} \mathbf{E} [m \langle \mathcal{P}X, X \rangle + \langle \mathcal{P}X, L_m \rangle - m \langle \mathcal{P}D_0, D_0 \rangle - \langle \mathcal{P}D_0, L_m \rangle].$$

We compute these terms for smooth m first. Compression may be omitted inside any of the displayed inner products, because both arguments belong to $\mathcal{H}$.

The identity term. Integration by parts and (1) give

$$\frac{1}{6} \mathbf{E} \left[m |\nabla Z|^2 + \sum_i m_i \langle \nabla_i Z, Z \rangle \right] = -\frac{1}{6} \mathbf{E} [m \langle \Delta Z, Z \rangle] = \mathbf{E} [m(-13e + 20B)].$$

This will be multiplied by the coefficient 4 in $\mathcal{P}$.

The derivative-slot curvature terms. Write $A_{ikjl} = (T_D)_{ikjl}$. The two terms to integrate are

$$\mathbf{E} [m A_{ikjl} \langle \nabla_i Z_j, \nabla_k Z_l \rangle + A_{ikjl} \langle \nabla_i Z_j, m_k Z_l \rangle].$$

Integrate the derivative k on the second paired tensor. The derivative of m cancels the second term exactly. Moreover $\nabla_k A_{ikjl} = 0$, by divergence freedom and skewness in i, k . The result is

$$-\mathbf{E} [m A_{ikjl} \langle \nabla_k \nabla_i Z_j, Z_l \rangle] = \frac{1}{2} \mathbf{E} [m A_{ikjl} \langle [\nabla_i, \nabla_k] Z_j, Z_l \rangle].$$

This calculation uses neither symmetry of X nor parallelness of T_D , and the factor $1/2$ comes only from skewing the two covariant derivatives.

The curvature in the derivative slot has the contraction

$$\frac{1}{2} (T_D)_{ikjl} R_{ikjh} = 3v^2 \delta_{lh}.$$

To see its normalization, substitute the quaternion expansions: the contraction over i, k removes all mixed plus/minus terms and gives $\frac{1}{2} \sum_s t_s (1 + t_s) \delta_{lh} = 3v^2 \delta_{lh}$. The matrix-slot curvature gives the quadratic form of

$$-\frac{1}{2} \sum_s t_s (1 + t_s) I_s \Sigma_s.$$

Since $t_s^2 = N_{ss} + 2v^2$, its compression is

$$-\frac{1}{2}(8Q_{5/2}(T) + 8Q_{5/2}(N) + 2v^2C) = 4v^2 - 4Q_{5/2}(T) - 4Q_{5/2}(N).$$

After division by 6, the full T_D contribution is therefore

$$\mathbf{E}[m(7v^2e - 4B - 4C_N)].$$

For U_D , the same integration gives $3\mathbf{E}[mb^2e]$. Its matrix-slot curvature contraction is zero, because the minus forms are orthogonal to the plus forms defining the curvature action on F .

The nonparallel matrix-slot coefficient. Put $A = B_F(T)$ temporarily. One integration by parts gives

$$\mathbf{E}[m\langle A\nabla_i Z, \nabla_i Z \rangle + m_i\langle A\nabla_i Z, Z \rangle] = -\mathbf{E}[m\langle A\Delta Z, Z \rangle] - \mathbf{E}[m\langle (\nabla_i A)\nabla_i Z, Z \rangle].$$

The operator $\nabla_i A$ is symmetric, so the last term is

$$\frac{1}{2}\mathbf{E}\left[\left\langle\left(m\Delta A + \sum_i m_i \nabla_i A\right)Z, Z\right\rangle\right].$$

Here $\nabla_i A = B_F(Z_i)$ and $\Delta A = 6B_F(T) - 6B_F(N)$, since B_F is a linear parallel construction. Also, pairing $B_F(T)Q_{5/2}(T)Z$ with Z gives $|Q_{5/2}(T)Z|^2$, because the latter vector belongs to G . After division by 6, the contribution is

$$\mathbf{E}\left[m(-10B - 3C_N + 20E_B) + \frac{1}{12}\sum_i m_i\langle B_F(Z_i)Z, Z \rangle\right].$$

Since $\langle B_F(Z_i)Z, Z \rangle = \text{tr}(Z_i \sum_j Z_j^2)$, multiplication by $-5/3$ produces exactly the middle term in (10). In particular, the variation of $B_F(T)$ has been retained explicitly; $\mathcal{P}$ is not assumed parallel.

The prescribed trace component. By Lemmas 5.1 and 5.2,

$$\mathcal{P}D_0 = \frac{5}{12}\text{Tr}^*(4 - B_F(T))\Delta T, \quad \text{Tr } L_m = \sum_i m_i Z_i.$$

The characteristic polynomial of a trace-free 3×3 matrix gives $B_F(T)T = N$ and $B_F(T)N = v^2T$. Thus

$$B_0 = (4 - B_F(T))(T - N) = (4 + v^2)T - 5N.$$

Using $\Delta T = 6(T - N)$, the two trace subtractions, before division by 6, are

$$-\frac{5}{2}\mathbf{E}\left[m\langle B_0, \Delta T \rangle + \sum_i m_i\langle B_0, Z_i \rangle\right] = \frac{5}{2}\mathbf{E}[m\langle \nabla B_0, Z \rangle].$$

Since $\nabla N = 2B_F(T)Z$ and $\langle T, Z_i \rangle = 6vp_i$, their normalized contribution is

$$\mathbf{E}\left[m\left(\left(10 + \frac{5}{2}v^2\right)e - 25B + 5v^2|p|^2\right)\right].$$

Multiplying the preceding contributions by their coefficients in $\mathcal{P}$ and collecting gives G_v as stated.

Finally, a global Lipschitz weight $m \geq m_0 > 0$ can be approximated uniformly and in H^1 by smooth positive weights with a common positive lower bound. The first derivatives converge in L^2 , so their quadratic products in the square cost converge in L^1 ; the uniformly bounded reciprocals of the weights pass to the same limit. All other terms use only the weight and its first derivative. Hence the identity and inequality extend to every weight in the stated class. $\square$

Lemma 5.4 (The quadratic-form matrices). *At a point where $v > 0$, use the three vector parameters p, k, l in (2). Then G_v is the quadratic form, with scalar entries multiplying vector dot products, whose matrix is*

$$\begin{pmatrix} -70 - 5b^2 + 100\sigma v - 25v^2 & -100\sqrt{3}\rho v & -35\rho v^2/\sqrt{3} \\ -100\sqrt{3}\rho v & -210 - 15b^2 - 300\sigma v - 190v^2 & -75v - 35\sigma v^2 \\ -35\rho v^2/\sqrt{3} & -75v - 35\sigma v^2 & -42 - 3b^2 - 10v^2 \end{pmatrix}. \quad (11)$$

Proof. Expanding the diagonal and off-diagonal matrix entries in (2) gives

$$\frac{B}{v} \longleftrightarrow \begin{pmatrix} 4\sigma/3 & -4\rho/\sqrt{3} & 0 \\ -4\rho/\sqrt{3} & -4\sigma & -1 \\ 0 & -1 & 0 \end{pmatrix}, \quad \frac{C_N}{v^2} \longleftrightarrow \begin{pmatrix} 4/3 & 0 & -\rho/\sqrt{3} \\ 0 & -4 & -\sigma \\ -\rho/\sqrt{3} & -\sigma & 0 \end{pmatrix}.$$

Here one uses $\sum s_i \tilde{s}_i = 0$, $\sum \tilde{s}_i^3 = -6\rho$, and $N/v^2 = \text{diag}(\sigma s_i + \rho \tilde{s}_i)$. The energy metric in these coordinates is $H = \text{diag}(5/3, 5, 1)$. If $\mathbf{B}$ denotes the matrix of B , the matrix of E_B is $\mathbf{B}H^{-1}\mathbf{B}$: $B_F(T)$ compressed to G is the self-adjoint operator whose quadratic form is B . Using $\sigma^2 + \rho^2 = 1$, this gives

$$\frac{E_B}{v^2} \longleftrightarrow \begin{pmatrix} 16/15 & 0 & 4\rho/(5\sqrt{3}) \\ 0 & 21/5 & 4\sigma/5 \\ 4\rho/(5\sqrt{3}) & 4\sigma/5 & 1/5 \end{pmatrix}.$$

Substitution into the definition of G_v gives (11) by matrix addition. The invariant expression for G_v is defined everywhere; on the zero set of T , all its derivative parameters vanish almost everywhere, as established for the norm identity. $\square$

5.3 The two costs of a variable weight

For the remaining argument choose the fixed positive weights

$$M = 1 - \frac{3}{8}(v - b), \quad m = M^2, \quad \overline{M} = 1 - \frac{3}{8}(b - v), \quad \overline{m} = \overline{M}^2. \quad (12)$$

They are one rational choice for the later coupling argument. Since $v + b < 2$, both M and $\overline{M}$ lie between $1/4$ and $7/4$. Almost everywhere,

$$dm = -\frac{3}{4}M(p - q), \quad d\overline{m} = -\frac{3}{4}\overline{M}(q - p).$$

Proposition 5.3 applies to these global Lipschitz weights. Their derivatives produce the two costs estimated next.

The Hessian has been eliminated from the integrated inequality. Two terms caused by the weight remain: a cubic contraction multiplied by dm , and a projected quadratic form in $dm \otimes Z$. We now bound both by first-derivative energies, while keeping the sign needed for the later scalar relaxation.

Lemma 5.5 (The cubic contraction). *For every real tangent vector a , with $Z_a = \sum_i a_i Z_i$,*

$$\left| \frac{\text{tr}(Z_a \sum_j Z_j^2)}{6} \right| \leq \frac{4}{5} \sqrt{\frac{3}{5}} |a| e^{3/2}. \quad (13)$$

Proof. By homogeneity, assume $|a| = 1$; the case $Z_a = 0$ is immediate. Put $\nu = |Z_a|/\sqrt{6}$, and diagonalize the normalized matrix $s = Z_a/\nu$. Apply the algebraic parametrization (2) with this s ,

rather than with T/v , and denote its vector parameters by $\hat{p}, \hat{k}, \hat{l}$. The diagonal entries of $Z_a = \nu s$ imply $\hat{p}_a = \nu$ and $\hat{k}_a = 0$. The part corresponding to the single component $\hat{p}_a a$ has energy $e_1 = 5\nu^2/3$. It is orthogonal in the energy metric to the remaining parameters. In the displayed matrix for B/v , it has no cross term with those parameters, because $\hat{k}_a = 0$ and the $\hat{p}, \hat{l}$ entry is zero. Its contribution to $\frac{1}{6} \sum_j \text{tr}(sZ_j^2)$ has absolute value at most $4e_1/5$.

For any symmetric trace-free 3×3 matrix S , its characteristic polynomial gives

$$|(S^2)_0|^2 = \frac{|S|^4}{6}.$$

Since $|s| = \sqrt{6}$, Cauchy–Schwarz yields $|\text{tr}(sS^2)| \leq |S|^2$. Applied to the remaining tensor, this bounds its contribution by $e - e_1$. Therefore

$$\left| \frac{\text{tr}(Z_a \sum_j Z_j^2)}{6} \right| \leq \nu \left(e - \frac{e_1}{5} \right) = \sqrt{\frac{3}{5}} e^{3/2} \sqrt{\frac{e_1}{e}} \left(1 - \frac{e_1}{5e} \right).$$

For $0 \leq r \leq 1$, the function $\sqrt{r}(1 - r/5)$ is increasing and has maximum $4/5$. This proves (13). $\square$

Lemma 5.6 (Real-vector projection identities). *Let P_+ be the projection in $\mathcal{H}$ onto total plus spin 3, retaining both possible minus spins. For any real tangent vector a and any $Z \in G$,*

$$|P_+(a \otimes Z)|^2 = \frac{7}{12} |a|^2 |Z|^2,$$

$$\langle \tilde{Q}P_+(a \otimes Z), P_+(a \otimes Z) \rangle = \frac{1}{2} |a|^2 \langle Q_{5/2}(T)Z, Z \rangle.$$

Proof. Complexify at the point, writing the outer tangent representation as $A \otimes B$, with A, B the plus and minus spin halves, and $G_{\mathbb{C}} = H_{5/2} \otimes B'$. For a real vector a , its reduced density on A is $\frac{1}{2}|a|^2 \text{Id}_A$. Indeed, the three traceless Hermitian operators on A are scalar multiples of $-iI_s$, and each has zero expectation because $a^\top I_s a = 0$. Its trace is $|a|^2$, which fixes the scalar part.

The complete vector is $a \otimes Z$, a product between the outer factor and the entire inner factor. Consequently, for every operator O acting only on $A \otimes H_{5/2}$,

$$\langle a \otimes Z, O(a \otimes Z) \rangle = \frac{|a|^2}{2} \langle Z, ((\text{Tr}_A O) \otimes \text{Id}_{B'})Z \rangle.$$

This partial trace retains every correlation of the inner plus factor with B' ; neither a nor Z is assumed to be a product within its own plus and minus factors.

The operator $\text{Tr}_A P_+$ is scalar on the irreducible space $H_{5/2}$. Its trace is the dimension 7 of spin 3, whereas $H_{5/2}$ has dimension 6. Thus it equals $7\text{Id}/6$, proving the first identity.

The family $\text{Tr}_A(P_+ \tilde{Q}(T) P_+)$ is equivariant and linear in the spin-2 variable T . Since spin 2 occurs once in $\text{End}(H_{5/2})$, it is a scalar multiple of $Q_{5/2}(T)$. To determine the scalar, take $T = \text{diag}(-1, -1, 2)$ and the inner highest weight $5/2$. The outer weight $1/2$ gives total weight 3 and $Q_3(T)$ -eigenvalue -1 . The outer weight $-1/2$ projects to total weight 2, where that eigenvalue is $(12 - 3 \cdot 2^2)/15 = 0$. The normalized outer trace is therefore $-1/2$. Since $Q_{5/2}(T)$ has eigenvalue -1 on its highest weight, the second identity has coefficient $1/2$, as asserted. $\square$

Proposition 5.7 (The projection cost). *For the weight (12),*

$$\frac{\langle \mathcal{P}(L_m)_Y, (L_m)_Y \rangle}{24m} \leq \frac{9}{64} |p - q|^2 \left[\frac{7}{12} (4 + y) e - \frac{5}{4} B \right]. \quad (14)$$

The bracket on the right is nonnegative. Both assertions remain valid under the closed inequalities $v + b \leq 2$ and $x + y \leq 2$.

Proof. Let P_- be the projection onto total minus spin 1, so the highest Hessian projection is P_+P_- . On that type, $-U_D \leq y$, and hence $\mathcal{P} \leq A$, where

$$A = 4 + y - \frac{5}{2}\tilde{Q}.$$

This operator acts only on the plus factors and commutes with P_- . On the entire high-plus subspace,

$$A \geq 4 + y - 2x \geq 0;$$

the last inequality follows from $x + y \leq 2$. It follows in operator order that $P_-AP_- \leq A$ there. Thus, for every real a ,

$$\langle \mathcal{P}(a \otimes Z)_Y, (a \otimes Z)_Y \rangle \leq \langle AP_+(a \otimes Z), P_+(a \otimes Z) \rangle = 6|a|^2 \left[\frac{7}{12}(4 + y)e - \frac{5}{4}B \right],$$

where the last equality is Lemma 5.6. This operator argument is valid for arbitrary correlations after the plus projection and does not discard a mixed minus-factor term.

Almost everywhere, $dm = -\frac{3}{4}M(p - q)$. Substituting $a = dm$, dividing by $24m$, and using $m = M^2$ gives (14). Finally, for any unit real a , the same equality and the lower bound for A give

$$\frac{7}{12}(4 + y)e - \frac{5}{4}B \geq \frac{7}{12}(4 + y - 2x)e \geq 0.$$

In particular, subsequent upper estimates may replace $|p - q|^2$ by $(|p| + |q|)^2$ in (14) without reversing the inequality. $\square$

6 Coupling the two Weyl blocks

Assume that neither Weyl block vanishes identically and that their second moments agree:

$$\mathbf{E}v^2 = \mathbf{E}b^2.$$

We retain the notation

$$\delta = 2 - v - b, \quad K = 4 + v^2 + b^2, \quad M = 1 - \frac{3}{8}(v - b), \quad m = M^2,$$

and place a bar over an expression to exchange the two blocks. In particular, $\bar{M} = 1 - \frac{3}{8}(b - v)$ and $\bar{m} = \bar{M}^2$.

The weights in (12) have opposite slopes along $v + b = 2$. Their derivatives introduce quartic terms in the first derivatives of curvature. We combine the two Hessian inequalities with one identity that controls those costs, the balanced variance defect, and tests of the weak norm equations. This produces a single integrand with nonnegative mean. The task is then to bound its quartic, quadratic, and potential parts by three expressions that complete one square.

6.1 The combined integral inequality

Start with the sum of the two integrands in (10). The three additions below are written explicitly in (16)–(18). Here $h \geq 0$ is a scalar coefficient and κ is a scalar multiplier, whose rational values are specified below. Their roles and integral signs are as follows; the degrees refer to first derivatives of curvature in the resulting integrand.

Added expression	Integral and derivative degrees	Purpose
$\Delta\delta$ identity	$\frac{h}{3}\mathbf{E}[\delta(Y ^2 + \bar{Y} ^2)] \geq 0$; degrees 4, 2, 0	Controls the quartic weight costs
Balanced variance defect	Nonnegative integral; degrees 2, 0	Controls $v-b$ using moment balance
κ norm tests	Zero integral; degrees 2, 0	Adjusts the quadratic and potential terms

We use the following verified rational choice for the constants and functions in this combination and its final comparison. It is one sufficient choice; no optimality or uniqueness is asserted. Set

$$D = 158, \quad h = \frac{17}{10}, \quad c = \frac{2}{5} - \frac{3}{10}vb, \quad a = v - 1, \quad s = b - 1, \quad n = \frac{\delta}{2}, \quad t = \frac{v-b}{2}.$$

Thus $c \geq 1/10$ on $v, b \geq 0, v + b \leq 2$. Define

$$\begin{aligned} \kappa(v, b) = & -\frac{D}{6}(a-s) + z_1 a^2 + \left(60 - \frac{2D}{9} + 2z_1\right)as + z_2 s^2 \\ & + \sum_{i=0}^3 z_{3+i} a^i s^{3-i} + \sum_{i=0}^4 z_{7+i} a^i s^{4-i}, \\ 4(z_1, \dots, z_{11}) = & (-43, -45, -63, 21, 183, 208, 61, 46, 101, 79, -41), \\ 4\Theta(v, b) = & n^2(1-n)(105+5n) + t^2(4+360n-586n^2) - 4t^4, \\ R_v = & \frac{4(1+v)}{1+6v+v^2}, \quad R_b = \frac{4(1+b)}{1+6b+b^2}. \end{aligned} \tag{15}$$

Here and below an exchanged expression is obtained by exchanging all block variables, and $\bar{\kappa} = \kappa(b, v)$. The multiplier κ changes the integrand through a zero-integral norm test. The functions Θ, c, R_v , and R_b enter only the comparison for the final square; they are not additional integrands. Neither κ nor Θ is required to have a sign.

The first addition is

$$\begin{aligned} h \left[(e + \bar{e}) \left(\lambda - \frac{e - |p|^2}{v} - \frac{\bar{e} - |q|^2}{b} \right) + 40\delta(B + \bar{B}) \right. \\ \left. - 30\delta \sum_{\pm} v^2(1 - 2\sigma v + v^2) \right], \end{aligned} \tag{16}$$

where

$$\lambda = 6(\sigma v^2 + \bar{\sigma} b^2 - v - b) - 26\delta.$$

The notation $\sum_{\pm}$ includes the displayed term and its exchange. To verify its sign, use

$$\Delta e = \frac{2}{6} |\nabla Z|^2 + 26e - 40B$$

and (3) in

$$\mathbf{E}[(e + \bar{e})\Delta\delta] = \mathbf{E}[\delta\Delta(e + \bar{e})].$$

The decomposition $\nabla Z = Y + D_0$ gives $|D_0|^2 = (5/12)|\Delta T|^2$. Moreover,

$$\frac{2|D_0|^2}{6} = 30v^2(1 - 2\sigma v + v^2).$$

Consequently the integral of (16) equals $\frac{h}{3}\mathbf{E}[\delta(|Y|^2 + |\bar{Y}|^2)]$, which is nonnegative.

The second addition is

$$D \left[\frac{|p - q|^2}{K} - \left(1 - \frac{\delta^2}{4} \right) (v - b)^2 \right]. \quad (17)$$

Indeed, the equality of the second moments implies

$$\mathbf{E}(v - b) = \frac{1}{2}\mathbf{E}[\delta(v - b)],$$

and hence

$$(\mathbf{E}(v - b))^2 \leq \frac{1}{4}\mathbf{E}[\delta^2(v - b)^2].$$

Applying (4) to $v - b$ proves that (17) has nonnegative integral. Finally add the zero-integral expression

$$\sum_{\pm} \left[\kappa(v, b) \left\{ 6v(1 - \sigma v) + \frac{e - |p|^2}{v} \right\} + \nabla \kappa(v, b) \cdot p \right]. \quad (18)$$

This is the weak identity (3) tested against $\kappa(v, b)$.

Let $\mathcal{I}$ denote the resulting total integrand. Thus $\mathbf{E}\mathcal{I} \geq 0$. All these integrals are legitimate without a positive lower bound for either norm: e/v and $\bar{e}/b$ are integrable by the regularization proving (3), while the energies and all scalar coefficients are bounded. Bounded Lipschitz test functions are admissible by smooth approximation, uniformly and in H^1 .

For the fixed weights in (12), dm is linear in p, q . Thus the cubic contraction multiplied by dm and the quadratic form in $L_m = dm \otimes Z$ have degree four in first derivatives of curvature. Grouping the exact total integrand by this degree gives $\mathcal{I} = \mathcal{I}_4 + \mathcal{I}_2 + \mathcal{I}_0$, where

$$\begin{aligned} \mathcal{I}_4 &= \sum_{\pm} \left[-\frac{5}{6} \sum_i m_i \frac{\text{tr}(Z_i \sum_j Z_j^2)}{6} + \frac{\langle \mathcal{P}(L_m)_Y, (L_m)_Y \rangle}{24m} \right] \\ &\quad - h(e + \bar{e}) \left(\frac{e - |p|^2}{v} + \frac{\bar{e} - |q|^2}{b} \right), \\ \mathcal{I}_2 &= \sum_{\pm} \left[mG_v + h\lambda e + 40h\delta B + \kappa(v, b) \frac{e - |p|^2}{v} + \nabla \kappa(v, b) \cdot p \right] + \frac{D}{K} |p - q|^2, \\ \mathcal{I}_0 &= -D \left(1 - \frac{\delta^2}{4} \right) (v - b)^2 \\ &\quad + \sum_{\pm} \left[6\kappa(v, b)v(1 - \sigma v) - 30h\delta v^2(1 - 2\sigma v + v^2) \right]. \end{aligned}$$

Here λ is the original symmetric expression in (16). The exchange in $\sum_{\pm}$ includes the operator, the Y -projection, the weight, the multiplier and its arguments, and all derivative labels. On a block's zero set, only its corresponding derivative summands vanish almost everywhere; the exchanged contributions remain. The quotients use the zero value assigned in (3), and G_v uses its invariant expression there. Products such as σv^2 use their continuous zero value; no continuity is asserted for the singular coefficients.

Proposition 6.1 (Three bounds for the combined integrand). *For this decomposition, define*

$$r = R_v|p|^2 + R_b|q|^2, \quad j = (|p|^2 + |q|^2) \left(\frac{|p|^2}{v} + \frac{|q|^2}{b} \right).$$

Then, almost everywhere,

$$\mathcal{I}_4 \leq -cj, \quad \mathcal{I}_2 \leq 2\Theta r, \quad \mathcal{I}_0 \leq -\frac{\Theta^2}{c}.$$

The corresponding derivative summands are assigned value zero on a zero set, as in (3). The total integrand and the bounds are integrable, and $\mathbf{EI} \geq 0$.

Section 8 combines these estimates with $r^2 \leq j$ to obtain

$$\mathcal{I} \leq -\frac{(\Theta - cr)^2}{c} - c(j - r^2).$$

The strictness needed for vanishing is proved there. We prove the three estimates in turn. The computations first take place on $v, b > 0$, $v + b \leq 2$, $x + y \leq 2$. The polynomial certificates used in the proof are supplied in Section 7.

6.2 The quartic terms

Put

$$e_0 = \frac{5}{3}|p|^2, \quad e_d = e - e_0 \geq 0, \quad z = |p| + |q|.$$

Since $dm = -\frac{3}{4}M(p - q)$, (13) bounds the cubic contraction multiplied by dm in (10) by $\frac{1}{2}\sqrt{3/5}Mze^{3/2}$. The nonnegative bracket in (14) is bounded above by $ge_0 + H_de_d$, where

$$g = \frac{7}{12}(4 + y) - \sigma v + \frac{\rho^2 v^2}{4}, \quad H_d = \frac{7}{12}(4 + y) + \frac{5}{4}v + 4.$$

In fact, the pure defect contribution to $-5B/4$ is at most $(5/4)ve_d$, and its mixed contribution is at most $2v|\rho|\sqrt{e_0e_d}$. The latter is at most $(\rho^2 v^2/4)e_0 + 4e_d$. Nonnegativity of the original bracket permits the replacement $|p - q|^2 \leq z^2$. We obtain

$$\begin{aligned} \mathcal{I}_4 \leq \sum_{\pm} \left[\frac{1}{2}\sqrt{\frac{3}{5}}zM e^{3/2} + \frac{9}{64}z^2(ge_0 + H_de_d) \right] \\ - h(e + \bar{e}) \left[\frac{(2/5)e_0 + e_d}{v} + \frac{(2/5)\bar{e}_0 + \bar{e}_d}{b} \right]. \end{aligned} \tag{19}$$

We first show that this upper bound decreases when either defect increases, with p, q fixed. Denote the right-hand side by $\mathcal{Q}$. For $e > 0$, direct differentiation gives

$$\begin{aligned} \partial_{e_d} \mathcal{Q} = \frac{3}{4}\sqrt{\frac{3}{5}}zM\sqrt{e} + \frac{9}{64}z^2H_d \\ - h \left[\frac{(2/5)e_0 + e_d}{v} + \frac{(2/5)\bar{e}_0 + \bar{e}_d}{b} + \frac{e + \bar{e}}{v} \right]. \end{aligned}$$

Set $u = \sqrt{\bar{e}/e}$. The bounds

$$\begin{aligned} z &\leq \sqrt{\frac{3}{5}}(\sqrt{e} + \sqrt{\bar{e}}), & M &\leq \frac{7}{4} - \frac{3}{4}v, & H_d &\leq \frac{15}{2} + \frac{2}{3}v, \\ \frac{2}{5}e_0 + e_d &\geq \frac{2}{5}e, & \frac{2}{5}\bar{e}_0 + \bar{e}_d &\geq \frac{2}{5}\bar{e}, & \frac{v}{b} &\geq \frac{v}{2-v} \end{aligned}$$

therefore imply

$$\frac{v}{e} \partial_{e_d} \mathcal{Q} \leq A_0(1+u)^2 + L_0(1+u) - h \left[\frac{7}{5} + \left(1 + \frac{2v}{5(2-v)} \right) u^2 \right],$$

where

$$A_0 = \frac{27}{320} v \left(\frac{15}{2} + \frac{2}{3} v \right), \quad L_0 = \frac{9}{20} v \left(\frac{7}{4} - \frac{3}{4} v \right).$$

All substitutions in the negative bracket use lower bounds; thus they give the required upper bound after multiplication by $-h$.

Write

$$C_0 = \frac{7}{5} h - A_0 - L_0, \quad D_0^{\text{sc}} = h - A_0 + \frac{2hv}{5(2-v)}.$$

Lemma 7.2 gives $C_0 > 0$ and

$$4C_0 \left((h - A_0)(2 - v) + \frac{2}{5} hv \right) - (2 - v)(2A_0 + L_0)^2 > 0.$$

For $0 < v < 2$ this says $4C_0 D_0^{\text{sc}} > (2A_0 + L_0)^2$; hence the homogenized quadratic

$$-C_0 + (2A_0 + L_0)u - D_0^{\text{sc}} u^2$$

is negative for every real u . No ordering of the two energies is needed. Before taking an endpoint limit, write the derivative bound as

$$\partial_{e_d} \mathcal{Q} \leq \frac{1}{v} \left[-C_0 e + (2A_0 + L_0) \sqrt{e\bar{e}} - D_0^{\text{sc}} \bar{e} \right].$$

It extends continuously to $e = 0$, where necessarily $e_0 = e_d = 0$. At $e = \bar{e} = 0$ its value is zero; otherwise the derivative remains negative. The exchanged argument applies to $\bar{e}_d$. Decreasing the defects one at a time consequently gives

$$\mathcal{Q}(e_0 + e_d, \bar{e}_0 + \bar{e}_d) \leq \mathcal{Q}(e_0, \bar{e}_0 + \bar{e}_d) \leq \mathcal{Q}(e_0, \bar{e}_0).$$

We may therefore put both defects equal to zero. Write

$$V = x - v, \quad W = y - b, \quad 0 \leq V \leq v, \quad 0 \leq W \leq b, \quad V + W \leq \delta.$$

We claim

$$g \leq g_v := \frac{7}{12} (4 + b + W) - v + 2V, \quad g_b = \frac{7}{12} (4 + v + V) - b + 2W.$$

For $r_0 = x/v \in [1, 2]$, use

$$\sigma = \frac{3r_0 - r_0^3}{2}, \quad 1 - \sigma = \frac{(r_0 - 1)^2(r_0 + 2)}{2}, \quad 1 + \sigma = \frac{(2 - r_0)(r_0 + 1)^2}{2}.$$

Since $r_0 v \leq 2$,

$$\frac{(r_0 - 1)(r_0 + 2)}{2} \left(1 + \frac{v(1 + \sigma)}{4} \right) \leq 2.$$

After substituting $v \leq 2/r_0$, the gap in this inequality is

$$\begin{aligned} & 2 - \frac{(r_0 - 1)(r_0 + 2)}{2} \left(1 + \frac{1 + \sigma}{2r_0} \right) \\ &= \frac{2 - r_0}{8r_0} \left[4r_0(r_0 + 3) - (r_0 - 1)(r_0 + 2)(r_0 + 1)^2 \right] \geq 0. \end{aligned}$$

The bracket is concave on $[1, 2]$ and has endpoint values 16 and 4. This also covers $r_0 = 2$ without division by $2 - r_0$. Multiplying the preceding bound by $V = v(r_0 - 1)$ proves the claim for g , using $\rho^2 = 1 - \sigma^2$.

Set $P = \sqrt{e_0}$ and $Q = \sqrt{\bar{e}_0}$. We will prove

$$\begin{aligned} & \left(\frac{67}{50} + \frac{27}{100}vb \right) (P^2 + Q^2)(bP^2 + vQ^2) \\ & \geq \frac{27}{128}vb(P + Q)^2(g_vP^2 + g_bQ^2) + \frac{3}{4}vb(P + Q)(MP^3 + \bar{M}Q^3). \end{aligned} \quad (20)$$

At zero defects, $z = \sqrt{3/5}(P + Q)$. Thus the positive terms of (19) are at most $2/5$ times the right-hand side of (20), divided by vb . Since

$$\frac{10}{9} \left(h - \frac{67}{50} - \frac{27}{100}vb \right) = c,$$

(20) gives the first estimate in Proposition 6.1; the factor $9/25$ that remains is exactly the conversion from $(e_0, \bar{e}_0)$ to $(|p|^2, |q|^2)$.

To prove (20), exchange the blocks, if necessary, so that $P > 0$ and $0 \leq u = Q/P \leq 1$; the case $P = Q = 0$ is immediate. In $g_v + g_bu^2$ the coefficients of V, W are nonnegative, and their difference is $(17/12)(1 - u^2) \geq 0$. Replacing (V, W) by $(\delta, 0)$ therefore enlarges this expression. This is a relaxation, not an assertion that the new pair is geometrically attained. The enlarged values are

$$g_v = \frac{19}{3} - 3v - \frac{17}{12}b, \quad g_b = \frac{7}{2} - \frac{19}{12}b.$$

Put $\omega = 27/128$, $\mu = 9/32$, and define

$$\begin{aligned} A &= \frac{67}{50}(1 + u^2), \\ G &= \frac{27}{100}(1 + u^2)u^2 + 3\omega(1 + u)^2 + \mu(1 + u)(1 - u^3), \\ H &= \frac{27}{100}(1 + u^2) + \frac{\omega}{12}(17 + 19u^2)(1 + u)^2 - \mu(1 + u)(1 - u^3), \\ L &= \omega(1 + u)^2 \left(\frac{19}{3} + \frac{7}{2}u^2 \right) + \frac{3}{4}(1 + u)(1 + u^3). \end{aligned}$$

Dividing the difference between the left and enlarged right sides of (20) by vbP^4 gives exactly

$$\frac{A}{v} + \frac{Au^2}{b} + Gv + Hb - L.$$

Here $A, L > 0$, and Lemma 7.2 gives $G, H > 0$. For

$$Z_* = L^2 - 4A(G + u^2H)$$

the same lemma gives $Z_* \leq 0$ on $[0, 1/16]$ and $Z_*^2 \leq 64A^2u^2GH$ on $[1/16, 1]$. In the latter interval, $Z_* \leq |Z_*| \leq 8Au\sqrt{GH}$. In both intervals it follows that

$$L \leq 2\sqrt{A}(\sqrt{G} + u\sqrt{H}).$$

Taking square roots is legitimate because both sides are nonnegative. The arithmetic–geometric mean inequalities applied to $A/v + Gv$ and $Au^2/b + Hb$ finish the proof of (20).

6.3 The quadratic terms

The quartic bound is established. We next bound the quadratic part by its *axial value*: the algebraic comparison with $\sigma = \bar{\sigma} = 1$, $\rho = \bar{\rho} = 0$, and $k = l = \bar{k} = \bar{l} = 0$, holding v, b, p, q fixed. The corresponding spectral shapes are $(-v, -v, 2v)$ and $(-b, -b, 2b)$. In the plus contribution, replacing $\bar{\sigma}$ by 1 raises the expression by $6h(1 - \bar{\sigma})b^2e \geq 0$. After this replacement, the shape-dependent quadratic expression is

$$mG_v + h\lambda_v e + 40h\delta B + \frac{\kappa}{v}(e - |p|^2) + (\partial_v \kappa)|p|^2 + (\partial_b \kappa)p \cdot q,$$

where $\lambda_v = 6(\sigma v^2 + b^2 - v - b) - 26\delta$. The last two terms and the separate variance term are unchanged by the shape comparison. Define

$$P_0 = 300m + 30hv + 160h\delta, \quad P_k = 300m + 160h\delta, \quad P_l = 35mv.$$

On the closed spectral triangle, $m \geq 1/16$ and $h, \delta, v \geq 0$; therefore $P_0 = P_k + 30hv > 0$. This supplies the positive denominator in the Schur comparison below. By (11) and (16), the loss in the pp entry when $\sigma < 1$ is $(1 - \sigma)vP_0/3$, and the mixed entries with (k, l) are $-v\rho(P_k, P_l)/\sqrt{3}$. The negative of the pure (k, l) matrix is exactly

$$D_\sigma = \begin{pmatrix} K_1 & K_{12} \\ K_{12} & K_2 \end{pmatrix} - \frac{\kappa}{v} \text{diag}(5, 1),$$

where

$$\begin{aligned} K_2 &= m(42 + 3b^2 + 10v^2) - h\lambda_v, \\ K_1 &= 5K_2 + m(300v\sigma + 140v^2) + 160h\delta v\sigma, \\ K_{12} &= m(75v + 35v^2\sigma) + 40h\delta v. \end{aligned}$$

Thus the required completion of squares follows from

$$P_0 \left[v \begin{pmatrix} K_1 & K_{12} \\ K_{12} & K_2 \end{pmatrix} - \kappa \text{diag}(5, 1) \right] - (1 + \sigma)v^2 \begin{pmatrix} P_k \\ P_l \end{pmatrix} \begin{pmatrix} P_k & P_l \end{pmatrix} \succeq 0. \quad (21)$$

Let A_σ be this matrix and $w = (P_k, P_l)^T$. The identity $P_0 = P_k + 30hv$ gives

$$-\partial_\sigma A_\sigma = v^2 \left[\begin{pmatrix} 30hv \\ -P_l \end{pmatrix} \begin{pmatrix} 30hv & -P_l \end{pmatrix} + \text{diag}(0, 6hvP_0) \right] \succeq 0.$$

Lemma 7.3 proves (21) at $\sigma = 1$, and this matrix monotonicity proves it for every $-1 \leq \sigma \leq 1$. When $\sigma < 1$, put $\ell = (1 - \sigma)vP_0/3 > 0$ and $d = -v\rho w/\sqrt{3}$. The Schur complement is

$$D_\sigma - \frac{dd^T}{\ell} = D_\sigma - \frac{(1 + \sigma)v}{P_0} ww^T = \frac{A_\sigma}{P_0 v} \succeq 0.$$

At $\sigma = 1$, one has $\rho = 0$ and

$$D_1 = \frac{A_1 + 2v^2 ww^T}{P_0 v} \succeq 0,$$

so no division by $1 - \sigma$ is needed. No sign of κ is assumed. Applying the exchanged argument proves the announced axial reduction.

At the axial comparison, (2) and the matrix for B/v give $e_{\text{ax}} = 5|p|^2/3$ and $B_{\text{ax}} = 4v|p|^2/3$. Put $\lambda_{\text{ax}} = 6(v^2 + b^2 - v - b) - 26\delta$. Since $\delta = -a - s$,

$$\begin{aligned} (h\lambda e + 40h\delta B)_{\text{ax}} &= \frac{5}{3}h(\lambda_{\text{ax}} + 32v\delta)|p|^2 \\ &= \frac{5}{3}h(6(a^2 + s^2) + 32a\delta)|p|^2. \end{aligned}$$

Together with the axial pp entry of (11), this gives $X_v|p|^2$ for the axial part of $mG_v + h\lambda e + 40h\delta B$, where

$$X_v = m(-70 + 100v - 25v^2 - 5b^2) + \frac{5}{3}h(6(a^2 + s^2) + 32a\delta), \quad X_b = X_v(b, v).$$

Collecting the remaining quadratic terms gives the desired estimate precisely when

$$\begin{aligned} \begin{pmatrix} \mathcal{Q}_v & J \\ J & \mathcal{Q}_b \end{pmatrix} &\preceq 2\Theta \text{diag}(R_v, R_b), \\ \mathcal{Q}_v &= X_v + \frac{D}{K} + \partial_v \kappa + \frac{2\kappa}{3v}, \quad \mathcal{Q}_b = X_b + \frac{D}{K} + \partial_b \bar{\kappa} + \frac{2\bar{\kappa}}{3b}, \\ J &= \frac{\partial_b \kappa + \partial_v \bar{\kappa}}{2} - \frac{D}{K}. \end{aligned} \tag{22}$$

The polynomial matrix certificate in Lemma 7.3 will prove this inequality.

The leading terms of κ can now be read in their intended role. Along the formal axial boundary $v = 1 + t$, $b = 1 - t$, direct substitution in the displayed formulas gives

$$\mathcal{Q}_v, \mathcal{Q}_b, J = O(t^2) \quad (t \rightarrow 0), \quad \Theta = t^2 - t^4.$$

Thus the constant and tangential linear terms cancel at the meeting point of the two axial blocks. This explains the local shape of the rational choice; the certificate proves its inequality on the full stated domain.

6.4 The potential and the zero sets

The quadratic estimate has been reduced to its explicit matrix certificate. It remains to bound the derivative-free potential and to extend the three inequalities across the zero sets. The potential is symmetric under exchanging the two blocks. For its pointwise estimate, suppose $v \geq b$ and define

$$\begin{aligned} E_0 &= D \left(1 - \frac{\delta^2}{4} \right) (v - b)^2 + 30h\delta(v^2 a^2 + b^2 s^2) + 6(av\kappa + sb\bar{\kappa}), \\ F &= cE_0 - \Theta^2, \quad L_v = 10h\delta v - \kappa, \quad L_b = 10h\delta b - \bar{\kappa}. \end{aligned}$$

Direct collection from (16)–(18) gives

$$\mathcal{I}_0 = -E_0 - 6v^2(1 - \sigma)L_v - 6b^2(1 - \bar{\sigma})L_b.$$

Since $y - b \leq \delta$ and $r_0 = y/b \leq 2$,

$$b^2(1 - \bar{\sigma}) = \frac{1}{2}b^2(r_0 - 1)^2(r_0 + 2) \leq 2\delta^2.$$

It therefore suffices to establish

$$L_v \geq 0, \quad F \geq 0, \quad F + 12c\delta^2 L_b \geq 0 \quad (v \geq b). \quad (23)$$

Indeed, if $L_b \geq 0$ then $\mathcal{I}_0 \leq -E_0 \leq -\Theta^2/c$. If $L_b < 0$, the displayed spectral bound instead gives

$$\mathcal{I}_0 \leq -E_0 - 12\delta^2 L_b \leq -\Theta^2/c.$$

Lemmas 7.3 and 7.4 prove all three inequalities in (23).

The use of an ordering in (22) and (23) is purely algebraic. All weights and their derivatives have already been formed from the fixed global labels of the two blocks. Their summed targets are symmetric. At a point with $v < b$, apply the ordered inequality to (b, v) and exchange the two vector entries. At $v = b$, either labeling gives the same conclusion. No maximum, minimum, or selected label is differentiated.

Finally, on the actual zero set $\{v = 0\}$, one has $x = 0$ and $y < 2$ by strict sectional positivity, and the components Z, p, k, l vanish almost everywhere. To pass the algebraic bounds to this set, first set all plus derivative parameters p, k, l to zero in the positive-variable inequalities. Then e, B, C_N and every numerator divided by v are identically zero before taking any limit. Keep $b > 0$ and all minus spectral and derivative data fixed. Choose axial plus spectra $(-v, -v, 2v)$, for which $x = v$, with $0 < v < \min\{2 - y, 2 - b\}$ tending to zero. This sequence remains in both required spectral inequalities. The remaining quadratic inequality is the minus diagonal, whose denominator $bK(1 + 6b + b^2)$ stays positive; the quartic bound becomes $-c|q|^4/b$. All cross terms with p vanish. This argument does not assert continuity of the singular plus coefficients. The other zero set is treated by exchange, and if both norms vanish all derivative components vanish. The potential inequalities are polynomial and extend continuously. The quartic quotients are bounded in absolute value by a constant times $e/v + \bar{e}/b$, and are therefore integrable. This completes the proof of Proposition 6.1, subject only to the explicit polynomial verifications in the next section.

7 Exact polynomial certificates

This section proves the finite polynomial inequalities used in Section 6. The coefficients are rational throughout. We give the conversion formulas, domains, and coefficient bounds so that each sign follows from a finite polynomial expansion. The method uses the nonnegative Bernstein basis on intervals and simplices; see Boudaoud–Caruso–Roy [5] and Roy [30, Sections 1 and 4]. We prove the conversion formulas and their matrix-valued consequence here. Positivity is verified for these specific polynomials, with their boundary zeros retained. The accompanying standard-library Python checker recomputes the saved exact coefficient data and checks the specified algebraic identities. Its coverage record distinguishes those executed reconstructions from the collection identities and geometric reductions proved in the text.

The source package accompanies this PDF under the stable article identifier

Positively-curved-Einstein-four-manifolds-September-23-2026

The arithmetic package is at `verification/`; from the paper directory, run

```
python3 -B verification/verify.py --check
```

The adjacent `COVERAGE.md` records the precise scope of that check. A detached copy of this PDF does not contain the checker.

7.1 Bernstein coefficients on intervals and triangles

Lemma 7.1 (Bernstein conversion). *Let p be a polynomial of degree at most N on an interval $[r, s]$, and write*

$$p(r + (s - r)z) = \sum_{j=0}^N a_j z^j.$$

Its degree- N Bernstein expansion is

$$p(r + (s - r)z) = \sum_{i=0}^N h_i \binom{N}{i} z^i (1 - z)^{N-i}, \quad h_i = \sum_{j=0}^i a_j \frac{\binom{i}{j}}{\binom{N}{j}}.$$

In particular, $h_i \geq \eta$ for all i implies $p \geq \eta$ on $[r, s]$.

For a polynomial on a triangle with ordered vertices $p_\ell = (r_\ell, s_\ell)$, set

$$V = r_3 + (r_1 - r_3)z + (r_2 - r_3)w, \quad W = s_3 + (s_1 - s_3)z + (s_2 - s_3)w.$$

If $P(z, w) = p(V, W) = \sum_{d,e} P_{de} z^d w^e$ has degree at most N , then its normalized degree- N triangular coefficients are

$$[p]_{ij} = s_{ij}(P) := \sum_{d=0}^i \sum_{e=0}^j \frac{\binom{i}{d} \binom{j}{e}}{\binom{N}{d+e} \binom{d+e}{d}} P_{de}, \quad i, j \geq 0, \quad i + j \leq N.$$

On $z, w \geq 0$, $z + w \leq 1$, the polynomial is the sum of these coefficients times the nonnegative basis functions

$$\frac{N!}{i!j!(N-i-j)!} z^i w^j (1 - z - w)^{N-i-j}.$$

These basis functions sum to one. The same expansion applies entrywise to a symmetric polynomial matrix: positive semidefiniteness of every coefficient matrix implies positive semidefiniteness throughout the triangle.

Proof. For an interval, expand $z^j = z^j(z + (1 - z))^{N-j}$ and compare coefficients in the Bernstein basis. On a triangle, expand

$$z^d w^e = z^d w^e (z + w + (1 - z - w))^{N-d-e}.$$

This gives the displayed factors after dividing each coefficient by the corresponding multinomial coefficient. Equivalently, substitute

$$(v, b) = \frac{u_1 p_1 + u_2 p_2 + u_3 p_3}{u_1 + u_2 + u_3}$$

and multiply by $(u_1 + u_2 + u_3)^N$. The normalized coefficient of $u_1^i u_2^j u_3^{N-i-j}$ is $[p]_{ij}$. The positivity statements follow from the nonnegative partition of unity, also for matrix quadratic forms. $\square$

7.2 The quartic scalar certificates

Lemma 7.2 (Scalar certificates). *For the polynomials in Section 6, the following bounds hold:*

$$\begin{aligned} C_0 &> 0, \quad 4C_0 \left((h - A_0)(2 - v) + \frac{2}{5}hv \right) - (2 - v)(2A_0 + L_0)^2 > 0 \quad (0 \leq v \leq 2), \\ G, H &> 0 \quad (0 \leq u \leq 1), \quad Z_* \leq 0 \quad (0 \leq u \leq 1/16), \\ Z_*^2 &\leq 64A^2 u^2 GH \quad (1/16 \leq u \leq 1). \end{aligned}$$

Proof. Apply the interval formula of Lemma 7.1 in the degrees and on the intervals indicated below. Every coefficient has at least the displayed positive lower bound. In the first two rows the bound applies on each of the two intervals separately.

Polynomial	Degree	Interval	Lower bound
C_0	2	$[0, 1], [1, 2]$	1/2
$4C_0((h - A_0)(2 - v) + \frac{2}{5}hv) - (2 - v)(2A_0 + L_0)^2$	5	$[0, 1], [1, 2]$	1/2
G	4	$[0, 1]$	9/10
H	4	$[0, 1]$	1/4
$-Z_*$	8	$[0, 1/16]$	13/100
$64A^2u^2GH - Z_*^2$	16	$[1/16, 1]$	12/100

For the first two rows the coefficients are in fact strictly greater than 1/2. For example, their respective minima on the two intervals are

$$\left(\frac{3971}{3200}, \frac{1063}{1600}\right), \quad \left(\frac{101027}{20000}, \frac{283729}{400000}\right).$$

All entries in this table are obtained by substituting the displayed formulas for A_0, L_0, C_0 and for A, G, H, L, Z_* into the coefficient sum in Lemma 7.1. Thus each bound is an inequality between rational coefficients. The lemma proves the required signs. $\square$

7.3 The two matrix certificates

For the quadratic inequality (22), introduce

$$\begin{aligned} r_v &= 1 + 6v + v^2, & r_b &= 1 + 6b + b^2, & \alpha &= \frac{5}{4}, \\ H_v &= 8vK\Theta(1 + v) - r_v \left\{ vD + K \left[v(X_v + \partial_v \kappa) + \frac{2\kappa}{3} \right] \right\}, \\ H_b &= H_v(b, v), \\ S_* &= 4(\alpha + \alpha^{-1})vb + \frac{\alpha b(1 - v)^2 + \alpha^{-1}v(1 - b)^2}{2}. \end{aligned}$$

The diagonal differences in (22) are $H_v/(vKr_v)$ and $H_b/(bKr_b)$. Since $r_v = 8v + (1 - v)^2$ and $r_b = 8b + (1 - b)^2$, weighted arithmetic-geometric mean gives

$$\sqrt{vbr_vr_b} \leq \frac{\alpha br_v + \alpha^{-1}vr_b}{2} = S_*.$$

It is therefore sufficient to prove

$$\begin{pmatrix} H_v & S_*KJ \\ S_*KJ & H_b \end{pmatrix} \succeq 0.$$

Indeed its diagonal entries are nonnegative and its determinant gives $H_vH_b \geq S_*^2K^2J^2 \geq vbr_vr_bK^2J^2$. Dividing by the positive denominators proves that the matrix of differences in (22) is positive semidefinite. This argument uses J^2 and imposes no sign condition on J .

For a symmetric 2×2 matrix, record its entries in the order (11, 22, 12) as a triple (r, h', s') . In this notation the two triples to be certified are

$$\begin{aligned} \mathcal{A} &= P_0(v(K_1, K_2, K_{12}) - \kappa(5, 1, 0)) - 2v^2(P_k^2, P_l^2, P_kP_l), & \sigma &= 1, \\ \mathcal{B} &= (H_v, H_b, S_*KJ). \end{aligned}$$

The first triple is precisely the matrix in (21) at $\sigma = 1$.

Lemma 7.3 (Matrix and boundary certificates). *The symmetric matrix represented by $\mathcal{A}$ is positive semidefinite on*

$$\mathcal{D} = \{(v, b) : v, b \geq 0, v + b \leq 2\}.$$

The symmetric matrix represented by $\mathcal{B}$ is positive semidefinite on

$$\mathcal{D}_+ = \{(v, b) \in \mathcal{D} : v \geq b\}.$$

On $\mathcal{D}_+$ one also has

$$L_v \geq 0, \quad F + 12c\delta^2 L_b \geq 0.$$

Proof. Use the vertices

$$\begin{aligned} O &= (0, 0), & S &= (1, 0), & C &= (1, 1), & U &= (2, 0), & W_0 &= (0, 2), \\ A &= (1/2, 1/2), & B &= (3/2, 1/2), & E &= (1, 1/2). \end{aligned}$$

A three-letter word specifies the vertices of an ordered triangle. The triangles OSC , OCW_0 , and SCU cover $\mathcal{D}$. The triangles OAS , ASE , ACE , SBE , CBE , and SUB cover $\mathcal{D}_+$; the latter is also the single triangle OCU , or the union $OSC \cup SCU$. All edges are included. The two subdivisions are shown below.

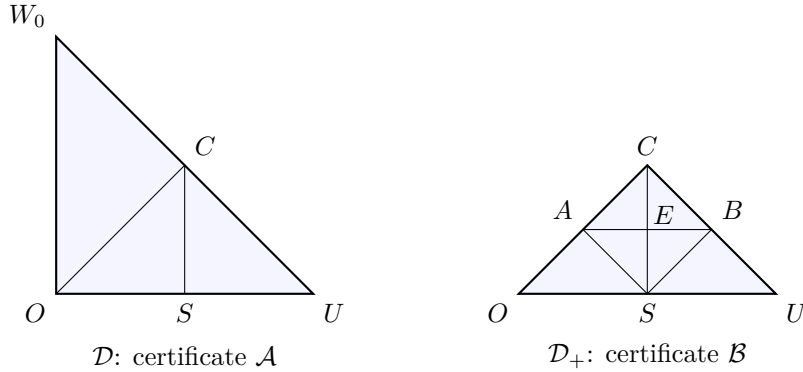


Figure 2: The full and ordered curvature domains, with the subdivisions used in the triangular coefficient table.

For a matrix row, the last two columns below bound, respectively,

$$\min([r]_{ij}, [h']_{ij}), \quad [r]_{ij}[h']_{ij} - [s']_{ij}^2.$$

Thus the last column is the determinant of each coefficient matrix. For a scalar row, the first bound applies directly to its coefficient. The bounds apply to every nonzero coefficient; a zero matrix exception means that all three of its entries vanish.

Object	Triangle	N	Zeros	Diagonal or scalar bound	Determinant bound
$\mathcal{A}$	OSC	8	0	16000	53000000
	OCW_0	8	0	890	3900000
	SCU	8	0	2400	3500000
$\mathcal{B}$	OAS	9	0	43	1050
	ASE	9	0	43	490
	ACE	10	1	2	21
	SBE	9	0	43	1300
	CBE	10	2	13/100	4
	SUB	9	0	5	290
L_v	OCU	6	1	9/10	—
$F + 12c\delta^2 L_b$	OSC	9	6	19/100	—
	SCU	9	7	9/1000	—

Here are explicit expansion instructions for all entries of the table. They also fix the coefficient ordering. Write $a = V - 1$, $s = W - 1$ in (15) and form $\kappa(V, W)$ and its two partial derivatives. A partial derivative is obtained by reducing the appropriate exponent of each monomial and multiplying by that exponent. Differentiate before substituting the affine expressions V, W . Put

$$\kappa_* = \kappa(V, W), \quad \bar{\kappa}_* = \kappa(W, V).$$

For clarity, the derivative of the exchanged polynomial with respect to its first variable is $(\partial_2 \kappa)(W, V)$, and its derivative with respect to its second variable is $(\partial_1 \kappa)(W, V)$.

Ordinary coefficient arrays are multiplied by convolution:

$$(PQ)_{de} = \sum_{f=0}^d \sum_{g=0}^e P_{fg} Q_{d-f, e-g}.$$

Use

$$\begin{aligned} m_* &= \left(1 - \frac{3}{8}(V - W)\right)^2, & d_0 &= 2 - V - W, \\ P_{k,*} &= 300m_* + 160hd_0, & P_{l,*} &= 35m_*V, \\ Y_* &= m_*(42 + 3W^2 + 10V^2) - h[6(V^2 + W^2 - V - W) - 26d_0]. \end{aligned}$$

For $\mathcal{A}$, apply s_{ij} from Lemma 7.1 to the three entries of

$$\begin{aligned} &(P_{k,*} + 30hV)\{V(A_1, Y_*, A_{12}) - \kappa_*(5, 1, 0)\} \\ &\quad - 2V^2(P_{k,*}^2, P_{l,*}^2, P_{k,*}P_{l,*}), \end{aligned}$$

where

$$A_1 = 5Y_* + m_*(300V + 140V^2) + 160hd_0V, \quad A_{12} = m_*(75V + 35V^2) + 40hd_0V.$$

These are the original entries of (21) at $\sigma = 1$, expressed only in the affine polynomials V, W .

For $\mathcal{B}$, put

$$\begin{aligned} K_* &= 4 + V^2 + W^2, & \Theta_* &= \Theta(V, W), \\ X_* &= m_*(-70 + 100V - 25V^2 - 5W^2) + \frac{5}{3}h(6(a^2 + s^2) + 32ad_0), \\ H_* &= 8VK_*\Theta_*(1 + V) \\ &\quad - (1 + 6V + V^2) \left\{ VD + K_* \left[V(X_* + (\partial_1 \kappa)(V, W)) + \frac{2\kappa_*}{3} \right] \right\}. \end{aligned}$$

Apply s_{ij} to H_* , to its exchange in V, W , and to

$$S_*(V, W) \left\{ \frac{K_*}{2} ((\partial_2 \kappa)(V, W) + (\partial_2 \kappa)(W, V)) - D \right\}.$$

For example, on CBE the triples at $(i, j) = (8, 0), (8, 1), (8, 2)$ are

$$\left(\frac{11041}{810}, \frac{4063}{810}, \frac{60557}{16200} \right), \quad \left(\frac{4328}{405}, \frac{839}{405}, \frac{3977}{2700} \right), \quad \left(\frac{191}{81}, \frac{191}{81}, -\frac{1189}{1350} \right).$$

For the two scalar rows, apply s_{ij} to $10hd_0V - \kappa_*$ and, respectively, to

$$c(V, W) \left\{ D \left(1 - \frac{d_0^2}{4} \right) (V - W)^2 + 30hd_0(V^2a^2 + W^2s^2) \right. \\ \left. + 6(Va\kappa_* + Ws\bar{\kappa}_*) + 12d_0^2(10hd_0W - \bar{\kappa}_*) \right\} - \Theta_*^2.$$

Together with the coefficient-sum formula, these prescriptions produce the displayed rational bounds by addition and multiplication of finite arrays.

The zero exceptions are as follows, with the third homogeneous index always $N - i - j$. For $\mathcal{B}$, they are the vertex C in ACE and CBE , namely $(0, 10, 0)$ and $(10, 0, 0)$, and also $(9, 1, 0)$ in CBE . For L_v the only zero is $(0, 6, 0)$, the vertex C in OCU . For $F + 12c\delta^2L_b$ on OSC , the zeros are exactly $i + j \leq 2$. On SCU they are exactly $9 - j \leq 2$, together with $(i, j) = (0, 6)$. There are no other exceptions.

Every nonzero matrix coefficient has positive diagonal entries and positive determinant, and each exceptional matrix is zero. Thus all coefficient matrices are positive semidefinite. The scalar coefficients are all nonnegative. Lemma 7.1 and the stated domain coverage prove the assertions. $\square$

7.4 The remaining potential polynomial

Lemma 7.4 (The polynomial F). *The polynomial $F = cE_0 - \Theta^2$ is nonnegative on $\mathcal{D}$.*

Proof. Use $v = 1 - n + t$, $b = 1 - n - t$. The full domain becomes

$$0 \leq n \leq 1, \quad |t| \leq 1 - n.$$

Write $F_*(n, t) = F(1 - n + t, 1 - n - t)$ for the pullback to these coordinates. It is even in t , and direct expansion gives $F_* = \sum_{j=0}^4 f_j(n)t^{2j}$, where

$$\begin{aligned} 240f_0 &= n^3(1 - n)(53511n^4 - 151593n^3 + 222537n^2 - 83607n + 10256), \\ 60f_1 &= n(19389n^5 - 418926n^4 + 523719n^3 - 289776n^2 + 67782n + 10564), \\ 60f_2 &= -1260129n^4 + 1370598n^3 - 451164n^2 + 23442n + 74, \\ 10f_3 &= -12200n^2 + 2601n + 75, \quad f_4 = -\frac{23}{5}. \end{aligned}$$

For $n < 1$, put $z = t^2/(1 - n)^2 \in [0, 1]$. The degree-four Bernstein coefficients in z are

$$h_i(n) = \sum_{j=0}^i \frac{\binom{i}{j}}{\binom{4}{j}} f_j(n) (1 - n)^{2j}, \quad 0 \leq i \leq 4.$$

Define

$$d_0(n) = n^3(1 - n), \quad d_1(n) = n(1 - n), \quad d_2(n) = d_3(n) = d_4(n) = 1 - n.$$

Each quotient $q_i = h_i/d_i$ is a polynomial. Applying the interval conversion formula to it on each of the four intervals

$$[0, 1/4], \quad [1/4, 1/2], \quad [1/2, 3/4], \quad [3/4, 1]$$

gives the following lower bounds for every coefficient:

i	Divisor d_i	Degree of q_i	Lower bound
0	$n^3(1 - n)$	4	3
1	$n(1 - n)$	6	31
2	$1 - n$	7	1/5
3	$1 - n$	7	12/5
4	$1 - n$	7	9/5

The complete prescription for each entry is to insert the displayed f_j into h_i , cancel the indicated polynomial factor, and apply the coefficient sum in Lemma 7.1 in the listed degree. In particular, the identities $h_i = d_i q_i$ hold as polynomial identities, including at the endpoints.

Since every d_i is nonnegative on $[0, 1]$, every h_i is nonnegative. Therefore

$$F_* = \sum_{i=0}^4 h_i(n) \binom{4}{i} z^i (1 - z)^{4-i} \geq 0 \quad (n < 1).$$

At $n = 0$, the coordinate $z = t^2$ still covers the entire permitted interval $|t| \leq 1$; the vanishing of h_0, h_1 causes no difficulty. At $n = 1$, the domain contains only $t = 0$, and directly $F_*(1, 0) = F(0, 0) = f_0(1) = 0$. Thus no numerical division by a vanishing factor is required, and all of $\mathcal{D}$ is covered. $\square$

Lemmas 7.2, 7.3, and 7.4 establish every polynomial inequality used in Section 6. Hence the three estimates of Proposition 6.1 hold on the stated positive-variable spectral domain. The compatible limit in Section 6 extends them to the zero sets of the standing strictly positive geometry.

8 Half-conformal flatness and the metric classification

The coupled estimates have been proved for the standing strictly positive geometry, including its zero sets. Their strictness now rules out two nonzero Weyl blocks. We then isolate the remaining harmonic alternative, exclude it by a conditional lower volume estimate, and identify the parallel curvature tensor. The orientation-cover argument comes only after the oriented conclusion has been established.

Proposition 8.1 (Half-conformal flatness). *At least one of T, U vanishes identically; equivalently, the metric is half-conformally flat.*

Proof. Suppose otherwise. Proposition 4.9 gives $\mathbf{E}v^2 = \mathbf{E}b^2$, so Proposition 6.1 applies. Its integrable total expression $\mathcal{I}$ has nonnegative integral and satisfies

$$\mathcal{I} \leq -cj + 2\Theta r - \Theta^2/c = -\frac{(\Theta - cr)^2}{c} - c(j - r^2), \quad c \geq \frac{1}{10},$$

where

$$r = R_v|p|^2 + R_b|q|^2, \quad j = (|p|^2 + |q|^2)(|p|^2/v + |q|^2/b).$$

The zero-set conventions established with (3) apply to the quotients. For $v > 0$,

$$1 + 6v + v^2 - 4\sqrt{v}(1 + v) = (\sqrt{v} - 1)^4,$$

so $R_v \leq 1/\sqrt{v}$, with equality only at $v = 1$. Cauchy–Schwarz therefore gives $r^2 \leq j$.

This inequality is strict almost everywhere on the set where p or q is nonzero. Indeed a Lipschitz function has zero gradient almost everywhere on each of its level sets: at a differentiability and density point of a level set, a nonzero differential would contradict density. Thus, outside a null set, an active gradient p has $v > 0$ and $v \neq 1$, and similarly for q . At such a point

$$r < |p|^2/\sqrt{v} + |q|^2/\sqrt{b} \leq \sqrt{j}.$$

Terms with vanishing gradients are read as zero. It follows that $\mathcal{I} \leq 0$ almost everywhere, and that $\mathcal{I} < 0$ almost everywhere where either gradient is nonzero. Since $\mathbf{E}\mathcal{I} \geq 0$, both gradients vanish almost everywhere. Connectedness and continuity make v, b constant. Both constants are positive, so the separate gaps (5) imply $v + b \geq 2$, contradicting $v + b \leq x + y < 2$. $\square$

Once one Weyl block vanishes, (M, g) belongs to the classically classified compact half-conformally flat Einstein class with positive scalar curvature [21, 15, 28]. We retain the lower volume and parallel curvature argument below as an independent proof of this final step under our standing hypothesis of strictly positive sectional curvature.

8.1 The remaining harmonic alternatives

After reversing orientation if necessary, one-block vanishing leaves $U = 0$. If T also vanishes, the sectional curvature is constantly one. For a nonzero surviving block, the characteristic identities reduce the possibilities to two integer values.

Lemma 8.2 (The one-block harmonic alternatives). *Suppose $U = 0$ and $T \neq 0$. Then*

$$\frac{V_M}{\pi^2} = \frac{8 - 2n_+}{3}, \quad s_0 = \mathbf{E}v^2 = \frac{3n_+}{4 - n_+}, \quad n_+ = 1 \text{ or } 2.$$

In the second case, $s_0 = 3$ and $V_M = 4\pi^2/3$.

Proof. Proposition 3.4 gives $n_- = 0$ and $\chi = 2 + n_+$, $\tau = n_+$. Solving its two identities yields the displayed volume and moment formulas. Positivity of volume gives $n_+ < 4$; the nonzero-block gap gives $s_0 \geq 1$, while $v < 2$ gives $s_0 < 4$. The finite nonnegative integer n_+ is therefore 1 or 2. Substitution gives the stated values in the second case. $\square$

8.2 Excluding the exceptional harmonic count

The value $n_+ = 2$ would force exactly $s_0 = 3$ and $V_M = 4\pi^2/3$. We now prove a strict lower bound under that same global moment hypothesis. The statement does not require one-block vanishing; that conclusion is used only to produce the exceptional value to which the bound will be applied.

Proposition 8.3 (Conditional lower volume bound). *If $s_0 = \mathbf{E}(v^2 + b^2) = 3$, then $V_M > 4\pi^2/3$.*

Recall from Section 4 that $\mathcal{E}$ is expectation over the unit tangent bundle, and that $\mathcal{R}$ is the normal Jacobi curvature matrix along its geodesics. With $\mathcal{J}'' + \mathcal{R}\mathcal{J} = 0$, $\mathcal{J}(0) = 0$ and $\mathcal{J}'(0) = \text{Id}$, put $D(t) = \det \mathcal{J}(t)$. We use the topology statement in Lemma 4.3, the geodesic averages in Lemma 4.2, and the discrete determinant in Lemma 4.5.

Lemma 8.4 (Lower polar bound). *One has*

$$V_M \geq 2\pi^2 \int_0^{\pi/\sqrt{3}} \mathcal{E}D(t) dt.$$

Proof. The open ball of radius $\pi/\sqrt{3}$ lies in the injectivity domain at every center by Lemma 4.3. Its volume is the corresponding integral of D and is at most V_M . Averaging over centers gives the bound. Increasing open radii justify the endpoint; it does not change the integral. $\square$

Lemma 8.5 (Lower radial comparison). *If $s_0 = 3$ and $0 < t < \pi/\sqrt{3}$, then, with $\theta = \sqrt{3}t$,*

$$\mathcal{E}D(t) \geq t^3 \left(\frac{\sin \theta}{\theta} \right)^{5/6} e^{-\theta^2/36}.$$

Moreover,

$$-\log \left(\frac{\sin \theta}{\theta} \right) - \frac{\theta^2}{6} \geq \frac{1}{2} \left(\frac{\theta}{\pi} \right)^4 + \frac{1}{3} \left(\frac{\theta}{\pi} \right)^6.$$

Proof. For each integer $N \geq 2$, put $a = t/N$. Let L_0 be the scalar Dirichlet tridiagonal matrix on the interior nodes $1, \dots, N-1$, with diagonal entries 2 and adjacent off-diagonal entries -1 . Put $G = a^2 L_0^{-1}$. Its entries are nonnegative; explicitly,

$$G_{ij} = a^2 \frac{\min(i, j)(N - \max(i, j))}{N}, \quad \text{tr } G = \frac{a^2(N^2 - 1)}{6} \rightarrow \frac{t^2}{6}.$$

Write $\mathcal{R}_N$ for the block diagonal matrix with blocks $\mathcal{R}(ia)$ and define $H_N = (G^{1/2} \otimes \text{Id}) \mathcal{R}_N (G^{1/2} \otimes \text{Id})$. For fixed $t < \pi/\sqrt{3}$ and large N ,

$$0 \prec H_N \preceq q_t \text{Id}, \quad q_t < 1, \quad \|H_N\| \leq \frac{3a^2}{2 - 2\cos(\pi/N)}, \quad \text{tr } H_N = 3 \text{tr } G.$$

In particular,

$$-\log \det(\text{Id} - H_N) = \sum_{k \geq 1} \frac{\text{tr } H_N^k}{k} \leq \frac{3 \text{tr } G}{1 - q_t}.$$

This bound is uniform over all geodesics and large N at the fixed time. It justifies taking expectations in the series and gives a uniform positive lower bound for the determinant ratios, independent of their increasing matrix dimensions.

By (8), $\mathcal{E}|\mathcal{R}|^2 = 3 + 3s_0/2 = 15/2$. For any ordered product of $k \geq 2$ curvature matrices,

$$|\text{tr}(\mathcal{R}_1 \cdots \mathcal{R}_k)| \leq 3^{k-2} |\mathcal{R}_1| |\mathcal{R}_k|.$$

Cauchy–Schwarz and invariance of geodesic-flow measure bound its expectation by $(5/6)3^k$. Neither commutation nor independence of the matrices is needed. Expanding the block trace of $((G \otimes \text{Id}) \mathcal{R}_N)^k$, its scalar coefficients are products of nonnegative entries of G . Their sum is $\text{tr}(G^k)$, so

$$\mathcal{E} \text{tr } H_N^k \leq \frac{5}{6} 3^k \text{tr}(G^k) \quad (k \geq 2).$$

The term $k = 1$ is exactly $3 \operatorname{tr} G$. Therefore

$$\mathcal{E} \log \det(\operatorname{Id} - H_N) \geq \frac{5}{6} \log \det(\operatorname{Id} - 3G) - \frac{1}{2} \operatorname{tr} G.$$

Jensen's inequality supplies a lower bound for the expected determinant by the exponential of this expression.

The flat block determinant is N^3 , and the normalization is exactly

$$\det(\operatorname{Id} - H_N) = \frac{\det \mathcal{L}_N}{N^3}.$$

Since $a = t/N$, Lemma 4.5 gives $\det(\operatorname{Id} - H_N) \rightarrow D(t)/t^3$, uniformly over geodesics at this fixed t : on the compact unit tangent bundle the curvature coefficients and their first derivatives are uniformly bounded. The scalar version gives

$$\det(\operatorname{Id} - 3G) \rightarrow \frac{\sin \theta}{\theta}.$$

Passing to these limits proves the first assertion. Equivalently, the uniform logarithmic bound permits passing averaged logarithms directly; applying Jensen before the limit already suffices.

Finally, in the scalar positive logarithmic series keep only the $k = 2, 3$ terms and, in each, the largest eigenvalue of G , which tends to t^2/π^2 . Subtracting $3 \operatorname{tr} G$ and passing to the limit yields the second assertion. $\square$

For the exact evaluation, use the new lower-comparison variable $u = \sqrt{3}t/\pi$. Lemma 4.7 gives $a_0 = 157/50 < \pi$ and

$$0 \leq S(\min(u, 1 - u)) \leq \sin(\pi u) \quad (0 \leq u \leq 1).$$

The two strict rational integral bounds below include the prefactor; the rows use their polynomial representatives up to the shared endpoint.

Integrand and prefactor	Interval	Degree	Lower bound
$\frac{a_0^3}{27} u^2(u^4/2 + u^6/3)S(\min(u, 1 - u))$	$[0, 1/2]$	15	$698/10^6$
same	$[1/2, 1]$	15	$40554/10^6$

They follow by the rational polynomial integration formula used in Section 4. Adding the base term $2(a_0^2 - 4)/9 = 4883/3750$ gives a total greater than $1343/1000 > 4/3$.

Proof of Proposition 8.3. Write the first bound in Lemma 8.5 relative to $t^3 \sin \theta/\theta$. Its exponential factor is

$$\exp \left[\frac{1}{6} \left(-\log \frac{\sin \theta}{\theta} - \frac{\theta^2}{6} \right) \right] \geq 1 + \frac{1}{6} \left[\frac{1}{2} \left(\frac{\theta}{\pi} \right)^4 + \frac{1}{3} \left(\frac{\theta}{\pi} \right)^6 \right].$$

Integrate Lemma 8.4, first up to any $T < \pi/\sqrt{3}$ and then let T increase to the endpoint. All integrands are nonnegative, so monotone convergence suffices; no uniform endpoint bound for a logarithm is required. The changes of variable $\theta = \sqrt{3}t$, $u = \theta/\pi$ give

$$\frac{V_M}{\pi^2} \geq \frac{2}{9}(\pi^2 - 4) + \frac{\pi^3}{27} \int_0^1 u^2(u^4/2 + u^6/3) \sin(\pi u) du.$$

The last two rational integral bounds and their constant term make this strictly greater than $1343/1000 > 4/3$, proving the assertion. $\square$

Lemma 8.2 makes the $n_+ = 2$ case equal to $V_M = 4\pi^2/3$ under $s_0 = 3$, contrary to Proposition 8.3. Hence only $n_+ = 1$ remains when exactly one block is nonzero.

8.3 Parallel curvature and global identification

Proposition 8.6 (The remaining parallel tensors). *Up to reversing orientation, either $T = U = 0$, or $U = 0$ and T is parallel with eigenvalues $2, -1, -1$.*

Proof. In the nonzero one-block case, the preceding exclusion gives $n_+ = 1$, so Lemma 8.2 gives $\mathbf{E}v^2 = 1$. The equality case of Proposition 3.3 yields $\nabla T = 0$ and $v = \sigma = 1$, hence eigenvalues $2, -1, -1$. The exchanged orientation is identical. If both blocks vanish, the curvature-operator decomposition gives constant sectional curvature one. $\square$

The model calculation is the Hopf-submersion curvature computation, following the framework of O'Neill [29]. We include it to fix the metric scaling precisely.

Lemma 8.7 (Curvature of the standard models). *The two tensors in Proposition 8.6 are the curvature tensors of the unit round sphere and of twice the Fubini–Study metric induced by the Hopf quotient of the unit sphere $S^5 \subset \mathbb{C}^3$.*

Proof. The first assertion follows immediately from the curvature-operator decomposition. In the second case choose at one point the orthogonal complex structure I corresponding to the eigenline of T with eigenvalue 2. On an orthonormal pair X, Y , the sectional curvature is

$$K(X, Y) = \frac{1}{2}(1 + 3\langle IX, Y \rangle^2).$$

Sectional curvatures determine an algebraic curvature tensor by polarization. Only this choice at a single tangent space is needed; no global complex structure is assumed.

For comparison, let g_H be the quotient metric on $S^5/S^1 = \mathbb{CP}^2$ obtained by isometric horizontal lift. At $z \in S^5$, the unit vertical field is iz . For horizontal lifts of base vector fields, the horizontal part of the sphere connection is the lifted base connection by the metric formula, while

$$(\nabla_X Y)^{\text{vert}} = -\langle iX, Y \rangle iz.$$

This follows by differentiating $\langle Y, iz \rangle = 0$. Substitution in the curvature formula yields

$$K_{g_H}(X, Y) = 1 + 3\langle iX, Y \rangle^2.$$

To track the factor three, the vertical bracket is $[X, Y]^{\text{vert}} = -2\langle iX, Y \rangle iz$ and invariant horizontal lifts satisfy $\nabla_{iz} Y = iY$. The differentiated vertical term and the vertical bracket therefore contribute, respectively, one and two copies of the same squared inner product. Equivalently, the extra horizontal term in the sphere curvature expression is $3\langle iX, Y \rangle iY$, whose pairing with X is $-3\langle iX, Y \rangle^2$ relative to the base expression. Scaling g_H by two gives the required sectional curvatures.

The quotient metric has parallel curvature: a unitary transformation that fixes a complex line and is minus the identity on its orthogonal complement induces an isometry fixing that point with tangent differential $-\text{Id}$. It acts by -1 on the covariant five-tensor ∇R , so $\nabla R = 0$. Transitivity gives this conclusion at every point. The round sphere likewise has parallel curvature. $\square$

Parallel curvature turns matching tensors at one point into matching Jacobi equations along paths. This is the continuation principle behind curvature-based isometry results such as Ambrose's [2]. We give the continuation and its inverse explicitly in the present compact, simply connected setting.

Lemma 8.8 (Global continuation of the model isometry). *The local identifications furnished by Lemma 8.7 extend to global isometries.*

Proof. Lemma 4.3 shows that M is simply connected. Both model spaces are simply connected as well. For the Hopf quotient this can be seen directly: local sections lift any loop to a path in S^5 . Join its endpoints in the circle fibre to close the path, contract the resulting loop in S^5 , and project the contraction. The added fibre path projects to a constant, so the original loop contracts.

Choose a tangent-space isometry matching the full curvature tensor at one point. On sufficiently small normal balls the exponential maps give an isometry: in parallel frames the Jacobi equations have identical coefficients because both curvature tensors are parallel. At a new point, the differential again matches curvature and defines the same construction on a new normal ball. Compactness supplies a common positive radius for these local constructions on the two manifolds.

An isometry germ is determined by its value and differential through exponential maps. Hence it continues uniquely along each path. Subdivide a homotopy square into small pieces lying in normal balls; uniqueness on overlaps shows that continuation is invariant under homotopy. Simple connectivity therefore gives a global local isometry. Continue the inverse germ in the same way on the model. The two compositions agree with the identity near the original points and then everywhere by continuation. They are inverse global isometries. $\square$

Corollary 8.9 (The oriented classification). *Every connected oriented smooth closed Einstein four-manifold with strictly positive sectional curvature is, after positive scaling, isometric to the round S^4 or to $\mathbb{CP}^2$ with its Fubini–Study metric.*

Proof. Strict sectional positivity gives a positive Einstein constant, so the metric can be normalized to $\text{Ric} = 3g$. Proposition 8.6 and Lemmas 8.7–8.8 identify that normalized metric with the unit round sphere or twice the Hopf quotient metric. Undoing the positive normalization proves the stated oriented conclusion. $\square$

8.4 The orientation cover and the third model

Completion of Theorem 1.1. Strict sectional positivity makes the Ricci tensor positive definite, so the Einstein constant is positive. If M is orientable, choose an orientation and apply Corollary 8.9.

Otherwise let $p : \widetilde{M} \rightarrow M$ be the orientation double cover. It is connected, closed and oriented; for the standard construction and connectedness see [20, Proposition 3.25]. The pulled-back metric $\tilde{g} = p^*g$ has the same Einstein and strict sectional-positivity properties. The nontrivial deck map τ switches the two local orientations over each point, so it is a free orientation-reversing involution, and $p \circ \tau = p$ makes it an isometry of $\tilde{g}$. After positive scaling, Corollary 8.9 identifies this cover with one of its two standard models. Conjugating τ by the model isometry preserves its freeness and its orientation sign.

The $\mathbb{CP}^2$ case is impossible. Its integral cohomology ring is $\mathbb{Z}[h]/(h^3)$ with $|h| = 2$, so h^2 generates $H^4(\mathbb{CP}^2; \mathbb{Z})$ [20, Theorem 3.19]. Any self-diffeomorphism sends h to $\pm h$ and therefore fixes h^2 . It has degree $+1$, contradicting the required orientation reversal.

On the unit round $S^4 \subset \mathbb{R}^5$, distance determines ambient inner products by $\langle x, y \rangle = \cos d(x, y)$. An isometry therefore preserves these inner products: its values on the five coordinate unit vectors determine an $A \in O(5)$, and comparison with their images gives $\tau(x) = Ax$ for every $x \in S^4$. Since $\tau^2 = \text{id}$, we have $A^2 = I$. An orthogonal involution has an orthogonal decomposition into its $+1$ and -1 eigenspaces. A nonzero $+1$ eigenvector would give a fixed point on the sphere, so freeness forces $A = -I$. The scaled base is consequently the round quotient $S^4/\{x \sim -x\} = \mathbb{RP}^4$.

The antipodal map has degree $(-1)^5 = -1$ on S^4 , so it reverses orientation [20, Section 2.2, property (f)]. Its quotient is a closed connected smooth manifold and inherits the Einstein and positive-sectional-curvature properties from the round sphere through the local isometry. Thus this third model does occur, and it is the only nonorientable case. $\square$

A Analytic foundations and characteristic formulas

This appendix proves the analytic inputs stated in Section 3. Throughout, M is the standing closed, connected, oriented four-manifold with $\text{Ric} = 3g$ and strictly positive sectional curvature. Only that normalized geometry and the conventions of Section 2 are used; none of the volume, Hessian, coupling, or classification conclusions enters.

A.1 Elliptic facts, density, and Poisson solvability

We use the following basic analytic facts on a closed smooth four-manifold. The inclusion $H^1 \subset L^4$ is continuous, and $H^1 \subset L^s$ is compact for $2 \leq s < 4$. Smooth scalar and bundle operators with Laplacian leading symbol satisfy the local $W^{2,p}$ estimates for $1 < p < \infty$, together with their iterated Sobolev consequences. All operators below have these hypotheses, and all nonlinear terms are put on the right side before applying the estimates. For scalar operators we use the strong interior estimates, Dirichlet solvability, and improvement of strong-solution integrability in Gilbarg–Trudinger [16, Theorem 9.11, pp. 235–236; Theorem 9.15 and Lemma 9.16, pp. 241–242]. The initial weak-to-strong passage for the nonlinear minimizer is justified in the next subsection, rather than inferred from an estimate for strong solutions. For a weak bundle equation with L^2 data, write the operator in a smooth local frame: its principal part is scalar, while its first-order coupling applied to an H^1 section belongs to L^2 . A cutoff reduces each component to a scalar equation for $\Delta_g - C$ on a smooth ball, with $C > 0$. Strong zero-boundary solvability and coercive energy uniqueness identify that component with its $W^{2,2}$ solution. Once this initial regularity is established, componentwise interior estimates and interpolation absorb the smooth first-order couplings; differentiation gives the higher regularity used for Poisson solutions and eigenfunctions.

Smooth functions are dense in $H^1(M)$: use local mollification in a finite coordinate cover and a smooth partition of unity. The same construction gives uniform and H^1 approximation for a Lipschitz function, as used in the weak norm testing rule.

For completeness, every smooth mean-zero function h has a unique smooth mean-zero solution of $\Delta\phi = h$. On the mean-zero subspace of H^1 , the Dirichlet energy is coercive. Indeed, otherwise a sequence with unit L^2 norm and gradients tending to zero has, by compactness in L^2 , a mean-zero limit with unit norm and zero weak gradient. On the connected manifold that limit is constant, a contradiction. The variational equation

$$\mathbf{E}\langle d\phi, d\psi \rangle = -\mathbf{E}(h\psi)$$

therefore has a unique mean-zero weak solution by the Hilbert-space representation theorem. The local regularity above makes it smooth. The same energy identity proves uniqueness. This is the Poisson fact used in Proposition 3.1.

A.2 The sharp Sobolev inequality

We give the normalized four-dimensional Einstein case of Ilias’s inequality [22, Theorem 3]. Its nonlinear Hessian proof belongs to the Bochner approach to elliptic rigidity; compare Bidaut–Véron–Véron [4].

Proof of Proposition 3.2. Fix $2 < s < 4$ and minimize

$$Q_s(w) = \frac{\mathbf{E}(w^2 + \frac{1}{2}|\nabla w|^2)}{(\mathbf{E}|w|^s)^{2/s}}$$

under $\mathbf{E}|w|^s = 1$. A minimizing sequence is bounded in H^1 ; weak compactness, compact inclusion into L^s , and lower semicontinuity give a minimizer. Taking its absolute value does not increase the energy, so choose $w \geq 0$. The ordinary Sobolev embedding makes the minimum m_s positive, and the constant test function gives $m_s \leq 1$. The Euler equation is

$$-\Delta w + 2w = 2m_s w^{s-1}.$$

After setting $u = m_s^{1/(s-2)} w$, it becomes

$$-\Delta u + 2u = 2u^{s-1}.$$

We record the regularity needed to use a negative power of u . We first pass from the weak minimizer to a strong solution. Choose a smooth coordinate ball B and a cutoff $\eta \in C_c^\infty(B)$. For $z = \eta u$ the weak equation gives

$$(\Delta_g - 2)z = -2\eta u^{s-1} + 2\langle d\eta, du \rangle + (\Delta_g \eta)u =: F \in L^{r_0}(B), \quad r_0 = \min\{2, 4/(s-1)\} > 4/3.$$

The strong Dirichlet existence theorem [16, Theorem 9.15] gives a solution $w_D \in W^{2,r_0}(B) \cap W_0^{1,r_0}(B)$ with this right side. The domain and coefficients are smooth, the operator is uniformly elliptic, and its zeroth-order coefficient is -2 . In dimension four, $r_0 > 4/3$ implies $W^{2,r_0} \subset W^{1,q}$ for some $q > 2$; compatibility of Sobolev traces therefore gives $w_D \in H_0^1(B)$. The difference $z - w_D \in H_0^1(B)$ solves $(\Delta_g - 2)(z - w_D) = 0$ weakly. Testing by $z - w_D$ shows that it vanishes. Thus u is locally W^{2,r_0} , and the strong interior estimates and integrability improvement now apply [16, Theorem 9.11 and Lemma 9.16].

Initially $u \in L^4$, while $u^{s-2} \in L^r$ with $r = 4/(s-2) > 2$. If $u \in L^q$, the equation has right side in L^p with $1/p = 1/q + 1/r$. The $W^{2,p}$ estimate and Sobolev embedding improve $1/q$ by a fixed positive amount smaller than $1/2 - 1/r$. Initially $p = 4/(s-1) > 1$. Iteration reaches $p > 2$, giving boundedness; at a borderline exponent $p = 2$, use any finite larger q before the next iteration.

After boundedness, the equation gives $u \in W^{2,p}$ for every finite p . Since $s-1 > 1$, the derivative of t^{s-1} is bounded on the bounded nonnegative range of u . Differentiating the equation once, and applying the same local regularity argument to each first coordinate derivative, gives $u \in W^{3,p}$ for every finite p . In particular, $u \in C^{2,\alpha}$ for some $\alpha > 0$. For $C \geq 2$ we have

$$(-\Delta_g + C)u = (C-2)u + 2u^{s-1} \geq 0.$$

The classical strong maximum principle gives $u > 0$, since $u \geq 0$ and $u \not\equiv 0$. A direct barrier proof uses a positive exponential radial function on an annulus tangent to the zero set: the weak maximum principle forces a nonzero inward derivative at the tangency point, contradicting nonnegativity and differentiability. Compactness now gives a positive minimum. The nonlinearity is smooth on this positive range, so further differentiation and elliptic regularity give smoothness of u .

Put $F = u^{-1}$. Its equation is

$$F\Delta F - 2|\nabla F|^2 + 2F^2 = 2F^{4-s}.$$

Since $\operatorname{div}(\operatorname{Hess}_0 F) = \frac{3}{4}d(\Delta F + 4F)$, differentiation gives

$$\begin{aligned} \operatorname{div}(F^{-3} \operatorname{Hess}_0 F) &= \frac{3}{4}F^{-4}[F d\Delta F + (\Delta F)dF - 4\operatorname{Hess} F(\nabla F, \cdot) + 4F dF] \\ &= \frac{3}{2}(4-s)F^{-1-s} dF. \end{aligned}$$

Pairing with dF and integrating yields

$$-\mathbf{E}F^{-3}|\text{Hess}_0 F|^2 = \frac{3}{2}(4-s)\mathbf{E}F^{-1-s}|dF|^2.$$

Both sides have opposite signs, so $dF = 0$. The positive constant solution of the equation for u is $u = 1$. The normalization of w then gives $w = 1$ and $m_s = 1$.

Thus the claimed inequality holds with 4 replaced by every $s \in (2, 4)$ and outer exponent $2/s$. For fixed $f \in H^1$, the ordinary inclusion $H^1 \subset L^4$ and dominated convergence give $\|f\|_{L^s(d\nu)} \rightarrow \|f\|_{L^4(d\nu)}$ as $s \uparrow 4$. This proves the stated critical inequality. $\square$

A.3 Harmonic forms and the characteristic identities

We derive the Euler and signature formulas using the complex of forms and an explicit signed heat computation. This is the heat-supertrace method of McKean–Singer [25]; general formulations and local coefficient calculations appear in Gilkey [17, Theorems 1.7.6 and 4.8.16].

We first record the spectral heat construction for a smooth symmetric Laplace-type operator $L = -\Delta + E$ on a Euclidean bundle over M . Adding a sufficiently large constant gives a coercive quadratic form on H^1 . Its weak inverse is compact and self-adjoint on L^2 by compact embedding, and elliptic regularity makes its eigenfunctions smooth. The spectrum is discrete with finite multiplicities and tends to infinity. Spectral heat evolution is smoothing for every positive time: polynomial powers of the shifted eigenvalue times $e^{-t\lambda}$ are bounded, and elliptic estimates for powers of L identify the corresponding Sobolev bounds.

For completeness, these estimates also justify smooth kernel and trace expansions. For a sufficiently large integer k , evaluation of any prescribed finite derivative after $(L + C)^{-k}$ is a bounded functional on L^2 , uniformly in its base point. Applied to an orthonormal eigenbasis, the Riesz representation theorem gives a uniform bound for the sum of the squared derivative values divided by $(\lambda + C)^{2k}$. Cauchy–Schwarz and the exponential factor then give uniform convergence of the spectral kernel series and all its derivatives for t in compact subintervals of $(0, \infty)$. Integrating the diagonal gives the usual heat trace. Uniqueness with prescribed initial point mass follows by testing the kernel against the eigenfunctions.

Lemma A.1 (The form-complex Laplacians). *Consider the elliptic complex*

$$\Lambda^0 \xrightarrow{d} \Lambda^1 \xrightarrow{\sqrt{2}d^+} \Lambda^+, \quad d^+ = P_+d.$$

Its three Laplacians are the Hodge Laplacians on the indicated bundles. Written as $-\Delta + E$, their potentials are

$$E_0 = 0, \quad E_1 = 3\text{Id}, \quad E_+ = 4\text{Id} - 2T.$$

Its alternating heat trace is $1 + n_+$.

Proof. The middle complex Laplacian is

$$dd^* + 2(d^+)^*d^+ = dd^* + d^*d + d^*\star d.$$

The last term is zero by $d^2 = 0$ and the formula for the codifferential. On self-dual two-forms, Hodge star exchanges dd^* and d^*d , so $2P_+dd^*$ is their sum. This identifies the last Laplacian; the scalar one is immediate. Their elliptic principal symbols are consequently those of the metric Laplacian.

Commuting covariant derivatives in the one-form Hodge Laplacian gives the Ricci endomorphism, equal to 3Id . For the plus potential, write a self-dual form as $\eta = \sum_s \eta_s I_s / \sqrt{2}$. Its Hodge energy is $2\mathbf{E}|d^*\eta|^2$, which equals

$$\mathbf{E} \left| \sum_s I_s \nabla \eta_s \right|^2 = \mathbf{E} \left(|\nabla \eta|^2 + \frac{1}{2} \langle C \nabla \eta, \nabla \eta \rangle \right),$$

where $C = \sum I_s \otimes \Sigma_s$ now uses the spin-one generators on the plus coefficients. The equality follows by expanding the quaternion products. Integrating the C term and taking its skew derivative commutator gives the potential

$$-\frac{1}{2} \sum_s (1 + t_s) \Sigma_s^2.$$

In the spin-one representation, $\Sigma_s^2 = 4(e_s e_s^\top - \text{Id})$. The potential is therefore $4\text{Id} - 2T$, as asserted.

The scalar kernel consists of the constants. A harmonic one-form satisfies

$$0 = \mathbf{E}(|\nabla \eta|^2 + 3|\eta|^2),$$

so the middle kernel vanishes. The last kernel has dimension n_+ . On each positive eigenspace of the complex Laplacian, the complex differential commutes with the Laplacian, and its adjoint divided by the eigenvalue is a contracting homotopy. That finite-dimensional complex has zero alternating dimension. Summing the heat factors over the positive spectrum leaves only the kernel contribution $1 + n_+$. $\square$

Lemma A.2 (Local signed heat coefficient). *For the three operators in Lemma A.1, let str denote the trace on Λ^0 minus that on Λ^1 plus that on Λ^+ . Write Ω for their connection curvatures. The constant local coefficient of their alternating heat trace, including the factor $(4\pi)^{-2}$, is*

$$\frac{1}{16\pi^2} \left(\frac{1}{2} \text{str } E^2 + \frac{1}{12} \text{str} \sum_{k,i} \Omega_{ki}^2 \right).$$

Proof. Fix the source point of the kernel. In normal coordinates z around it, use radially parallel orthonormal frames on each of the three bundles and set $\mu = \sqrt{\det g}$, $h = \mu^{-1/2}$. The ansatz

$$(4\pi t)^{-2} e^{-|z|^2/(4t)} \sum_{j \geq 0} t^j U_j(z)$$

gives $U_0 = h\text{Id}$ and the transport equations

$$\left(z \cdot \partial + j + \frac{1}{2} z \cdot \partial \log \mu \right) U_j = -L U_{j-1}.$$

Here the radial connection coefficient is zero. The scalar part follows also from $\Delta|z|^2 = 8 + 2z \cdot \partial \log \mu$. Solving the radial equation gives

$$U_j(z) = -h(z) \int_0^1 s^{j-1} (h^{-1} L U_{j-1})(sz) ds.$$

These transport solutions compute the actual local asymptotics. Indeed, truncate at any large N and apply a cutoff inside a normal ball. The resulting kernel has initial delta mass, and the transport equations leave a residual uniformly $O(t^{N-2})$; derivatives of the cutoff give exponentially small terms. The heat evolution of a metric connection with bounded potential has a bounded sup norm on bounded time intervals, by the maximum principle for the squared norm. Duhamel

integration therefore makes the difference from the true kernel $O(t^{N-1})$. Compactness permits a uniform normal-ball radius and uniform constants in the source point. Taking N large justifies the diagonal coefficient calculation below.

In these frames write the connection as $\partial_i + A_i$. Every A_i is skew-symmetric and radial gauge gives

$$A_i(z) = \frac{1}{2} z^k \Omega_{ki}(0) + O(|z|^2).$$

For example, the exact radial-gauge identity $A_i(z) = \int_0^1 s z^k \Omega_{ki}(sz) ds$ follows by contracting the curvature equation with z^k and integrating along the radius. The signed ranks and potentials satisfy, at every point,

$$\text{str Id} = 1 - 4 + 3 = 0, \quad \text{str } E = -12 + \text{tr}(4 \text{Id} - 2T) = 0.$$

In coordinate form,

$$L = -g^{ij}[(\partial_i + A_i)(\partial_j + A_j) - \Gamma_{ij}^k(\partial_k + A_k)] + E.$$

Apply this to $U_0 = h \text{Id}$. All purely scalar terms have zero supertrace. Every term linear in A_i or a coordinate derivative of A_i has zero ordinary trace in each bundle, by skewness. The potential has zero supertrace pointwise. Thus the exact surviving expression is

$$\text{str}(LU_0)(z) = -h(z)g^{ij}(z)\text{str}(A_i(z)A_j(z)).$$

In particular, derivatives of the varying potential have already disappeared before the radial integral is taken. The transport formula now gives

$$\begin{aligned} \text{str } U_1(z) &= h(z) \int_0^1 g^{ij}(sz) \text{str}(A_i(sz)A_j(sz)) ds \\ &= \frac{1}{12} z^k z^\ell \text{str} \sum_i \Omega_{ki}(0) \Omega_{\ell i}(0) + O(|z|^3). \end{aligned}$$

The coefficient is $\frac{1}{4} \int_0^1 s^2 ds = \frac{1}{12}$. Scalar and metric corrections do not change this quadratic term.

At the center, $A_i(0) = 0$, $\Gamma(0) = 0$, and $\sum_i \partial_i A_i(0) = \frac{1}{2} \sum_i \Omega_{ii}(0) = 0$ as an endomorphism. Consequently

$$U_1(0) = c \text{Id} - E(0), \quad (LU_1)(0) = -\sum_i \partial_i^2 U_1(0) + E(0)U_1(0),$$

where c is the same scalar for all three bundles. Thus there are no additional connection terms at the center. Taking the Laplacian of the signed quadratic expression gives

$$\text{str} \sum_i \partial_i^2 U_1(0) = \frac{1}{6} \text{str} \sum_{ki} \Omega_{ki}^2, \quad \text{str}(EU_1(0)) = -\text{str } E^2.$$

Since the transport equation at the center gives $U_2(0) = -\frac{1}{2}(LU_1)(0)$, we obtain

$$\text{str } U_2(0) = \frac{1}{2} \text{str } E^2 + \frac{1}{12} \text{str} \sum_{ki} \Omega_{ki}^2.$$

The lower coefficients have zero signed diagonal trace. Multiplication by the heat prefactor $(4\pi t)^{-2}$ now proves the assertion. $\square$

Proof of Proposition 3.4. By Hodge theory, harmonic forms represent de Rham cohomology. The scalar kernel is one-dimensional, and the one-form kernel vanishes by Lemma A.1. Hodge star gives the corresponding dimensions in degrees four and three. On harmonic two-forms, the intersection form is positive definite on the self-dual part and negative definite on the anti-self-dual part. Thus its signature is $n_+ - n_-$, and the alternating sum of the Betti numbers is $2 + n_+ + n_-$. This identifies the integers in Proposition 3.4 with the topological signature and Euler characteristic without using simple connectivity.

The potential-square supertrace is

$$\text{str } E^2 = -\text{tr}(9\text{Id}_{\Lambda^1}) + \text{tr}((4\text{Id}_{\Lambda^+} - 2T)^2) = 12 + 24v^2.$$

On one-forms, the connection curvature has plus and minus generators I_s, J_s . Orthogonality of the two-form bases, together with $I_s^2 = J_s^2 = -\text{Id}$, gives

$$\text{tr}_{\Lambda^1} \sum_{ki} \Omega_{ki}^2 = -4 \sum_s ((1 + t_s)^2 + (1 + u_s)^2) = -24(1 + v^2 + b^2).$$

On plus two-forms only the plus generators act, and each speed-two spin-one generator has square trace -8 . Therefore

$$\text{tr}_{\Lambda^+} \sum_{ki} \Omega_{ki}^2 = -8 \sum_s (1 + t_s)^2 = -24(1 + 2v^2).$$

The scalar connection is flat. Hence $\text{str} \sum \Omega_{ki}^2 = 24(b^2 - v^2)$, and Lemma A.2 gives the signed constant density

$$\frac{1}{16\pi^2}(6 + 10v^2 + 2b^2).$$

Integrate this against $d\text{vol}_g$ and compare with the constant alternating heat trace from Lemma A.1:

$$16\pi^2(1 + n_+) = \text{Vol}(M) \mathbf{E}(6 + 10v^2 + 2b^2).$$

Reverse orientation to get the analogous formula with n_-, b, v . Adding and subtracting the two formulas gives exactly (6).

On anti-self-dual forms the potential is $4\text{Id} - 2U$. If $U = 0$, the energy of a harmonic anti-self-dual form is the integral of $|\nabla\eta|^2 + 4|\eta|^2$, so its kernel is zero. The assertion for $T = 0$ follows in the same way. $\square$

B Minimizing geodesics and topology

We prove Lemma 4.3 in the standing setting: M is a closed, connected, oriented smooth four-manifold with $\text{Ric} = 3g$ and strictly positive sectional curvature. Along a unit-speed geodesic, let $\mathcal{R}$ be the normal Jacobi curvature matrix in a parallel orthonormal frame. The hypotheses give

$$0 < \mathcal{R} < 3\text{Id}, \quad \text{tr } \mathcal{R} = 3.$$

The trace supplies the diameter estimate, the upper bound supplies the conjugate-radius estimate, and strict positivity supplies the shortening variations. Orientation is used in the normal-holonomy argument. These are the classical methods of Myers, Synge, and Klingenberg [27, 31, 23]; the proof below retains the endpoint variations and proves simple connectivity independently of all volume and classification conclusions.

Proof of Lemma 4.3. For a unit-speed geodesic of length L , the index form on normal fields with zero endpoint values is

$$I(Y, Y) = \int_0^L (|Y'|^2 - \langle \mathcal{R}Y, Y \rangle) dt.$$

It is nonnegative on a minimizing segment, by the second variation of energy. Summing it on three parallel normal unit fields multiplied by $\sin(\pi t/L)$ gives

$$\frac{3L}{2} \left(\frac{\pi^2}{L^2} - 1 \right),$$

so a minimizing segment has length at most π . It cannot contain an interior conjugate point: a nonzero Jacobi field vanishing at the initial point and that interior point, extended by zero, has index zero but is not in the nullspace of the index form. The latter assertion follows by integration by parts against a field whose value pairs nontrivially with the jump in its derivative. A nonnegative quadratic form, however, pairs a zero-energy vector to zero against every vector, a contradiction.

The ordinary Dirichlet Poincaré inequality and $\mathcal{R} < 3\text{Id}$ show that a nonzero field with zero endpoints has positive index when $L \leq \pi/\sqrt{3}$. Thus no conjugate point occurs at such a distance. We show that no cut point occurs at a smaller distance either. Let c be the infimum of all cut radii; compactness and normal balls give $c > 0$. Suppose $c < \pi/\sqrt{3}$ and choose cut segments with lengths tending to c , all below the conjugate-radius bound. Each endpoint has two distinct minimizing segments from its starting point. To see this, extend a cut segment a little and take limits of minimizers to the extended endpoints. If the limiting minimizer were the original segment alone, local invertibility of the exponential map would force the extended segment to remain minimizing, contrary to its cut time.

After passing to a subsequence, compactness gives two length- c minimizing segments with common endpoints p, q . Their initial vectors cannot coalesce: the endpoint map

$$(p, w) \mapsto (p, \exp_p w)$$

is a local diffeomorphism at every vector in question, so two converging branches in the same inverse-function neighborhood would coincide. This map also supplies two distinct smooth local geodesic branches for nearby pairs of endpoints.

Their terminal unit velocities must be opposite. If these velocities are u_1, u_2 with $u_2 \neq -u_1$, move q initially in direction $-(u_1 + u_2)$. The first length variation of both branches is $-(1 + \langle u_1, u_2 \rangle) < 0$. Both branches would then have length less than c . Every geodesic of length less than the global cut infimum is minimizing and lies in the injectivity domain, so two distinct such branches are impossible. Moving p instead proves that their initial velocities are opposite as well. The first segment followed by the reverse of the second is therefore a smooth closed geodesic.

Parallel transport around this closed geodesic fixes the tangent and preserves orientation. Its action on the three-dimensional normal space belongs to $\text{SO}(3)$, and hence fixes a nonzero vector. Let V be the resulting periodic parallel normal field. Move both endpoints along geodesics with initial velocities $V(p), V(q)$, so their accelerations at the variation origin vanish. The local geodesic branches vary smoothly. Their variation fields J are normal: their tangential components are affine and have zero boundary values. For either branch, $W = V - J$ has zero boundary values and $I(J, W) = 0$, while $I(W, W) \geq 0$ because the unvaried segment is minimizing. Thus

$$I(J, J) \leq I(V, V) = - \int \langle \mathcal{R}V, V \rangle < 0.$$

The first length variations vanish, and these negative index values are the second length variations, since there are no endpoint acceleration or tangential terms. Both branch lengths therefore become

less than c . They remain distinct in their disjoint inverse-function neighborhoods, and lengths below c force them to be minimizing. This is the same contradiction as above, proving the injectivity bound.

Finally, a nontrivial free homotopy class would have a shortest representative. Indeed, constant-speed minimizing sequences have a uniform Lipschitz bound; compactness gives a uniformly convergent subsequence, and sufficiently close loops are freely homotopic by the short normal geodesics between corresponding points. A uniform normal ball radius also gives a positive lower bound on lengths of nontrivial loops. Local geodesic replacement shows that the minimizing loop is a nonconstant smooth closed geodesic. Its normal holonomy again fixes a nonzero parallel field, whose periodic variation has strictly negative second length variation. The variation preserves its free homotopy class, a contradiction. Hence the fundamental group is trivial. $\square$

References

- [1] U. Abresch and W. T. Meyer, *Injectivity Radius Estimates and Sphere Theorems*, in *Comparison Geometry*, MSRI Publications **30**, Cambridge University Press, 1997, 1–47. [Publisher-hosted text](#).
- [2] W. Ambrose, *Parallel Translation of Riemannian Curvature*, Ann. of Math. (2) **64** (1956), no. 2, 337–363. [doi:10.2307/1969978](#).
- [3] M. Berger, *Sur quelques variétés d'Einstein compactes*, Ann. Mat. Pura Appl. **53** (1961), 89–95. [doi:10.1007/BF02417787](#).
- [4] M.-F. Bidaut-Véron and L. Véron, *Nonlinear elliptic equations on compact Riemannian manifolds and asymptotics of Emden equations*, Invent. Math. **106** (1991), 489–539. [doi:10.1007/BF01243922](#). Erratum, Invent. Math. **112** (1993), no. 2, 445. [doi:10.1007/BF01232442](#).
- [5] F. Boudaoud, F. Caruso, and M.-F. Roy, *Certificates of Positivity in the Bernstein Basis*, Discrete Comput. Geom. **39** (2008), no. 4, 639–655. [doi:10.1007/s00454-007-9042-x](#).
- [6] S. Brendle, *Einstein manifolds with nonnegative isotropic curvature are locally symmetric*, Duke Math. J. **151** (2010), no. 1, 1–21. [doi:10.1215/00127094-2009-061](#).
- [7] D. M. J. Calderbank, P. Gauduchon, and M. Herzlich, *Refined Kato inequalities and conformal weights in Riemannian geometry*, J. Funct. Anal. **173** (2000), no. 1, 214–255. [doi:10.1006/jfan.2000.3563](#).
- [8] X. Cao and H. Tran, *Einstein four-manifolds of pinched sectional curvature*, Adv. Math. **335** (2018), 322–342. [doi:10.1016/j.aim.2018.07.003](#). Author version: [arXiv:1612.06023v2](#).
- [9] G. Catino and D. Dameno, *Conformal Kähler rigidity of Einstein four-manifolds*, 2026, [arXiv:2607.21538v1](#) [math.DG].
- [10] L. Cheng, *On Positively Curved Einstein Four-Manifolds with Euler Characteristic at Most Three*, [arXiv:2609.23337v1](#) [math.DG], September 20, 2026.
- [11] E. Costa, *On Einstein four-manifolds*, J. Geom. Phys. **51** (2004), no. 2, 244–255. [doi:10.1016/j.geomphys.2003.10.013](#).
- [12] D. Dameno, *A note on Einstein metrics and Riemannian twistor spaces*, Math. Z. **313** (2026), article 86. [doi:10.1007/s00209-026-04100-w](#).
- [13] A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. **49** (1983), no. 3, 405–433. [Journal archive](#).
- [14] L. F. Di Cerbo, *A rigidity theorem for Einstein 4-manifolds with sectional curvature of a fixed sign, and its consequences*, Math. Ann. **396** (2026), article 11. [doi:10.1007/s00208-026-03547-w](#). Author version: [arXiv:2503.09570v3](#).
- [15] Th. Friedrich and H. Kurke, *Compact four-dimensional self-dual Einstein manifolds with positive scalar curvature*, Math. Nachr. **106** (1982), no. 1, 271–299. [doi:10.1002/mana.19821060124](#).

- [16] D. Gilbarg and N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, Classics in Mathematics, Springer, Berlin, 2001, reprint of the 1998 edition. doi:10.1007/978-3-642-61798-0.
- [17] P. B. Gilkey, *Invariance Theory, the Heat Equation, and the Atiyah–Singer Index Theorem*, Publish or Perish, 1984. Author-hosted electronic reprint, 1996.
- [18] M. J. Gursky and C. LeBrun, *On Einstein manifolds of positive sectional curvature*, Ann. Global Anal. Geom. **17** (1999), no. 4, 315–328. doi:10.1023/A:1006597912184.
- [19] M. J. Gursky and A. Malchiodi, *Einstein four-manifolds of positive sectional curvature and ADM mass*, arXiv:2609.30803v1 [math.DG], submitted September 25, 2026.
- [20] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002. Author-hosted text.
- [21] N. J. Hitchin, *Kählerian twistor spaces*, Proc. London Math. Soc. (3) **43** (1981), no. 1, 133–150. doi:10.1112/plms/s3-43.1.133.
- [22] S. Ilias, *Constantes explicites pour les inégalités de Sobolev sur les variétés riemanniennes compactes*, Ann. Inst. Fourier (Grenoble) **33** (1983), no. 2, 151–165. doi:10.5802/aif.921.
- [23] W. Klingenberg, *Contributions to Riemannian Geometry in the Large*, Ann. of Math. (2) **69** (1959), no. 3, 654–666. doi:10.2307/1970029.
- [24] C. Koca, *Einstein Hermitian metrics of positive sectional curvature*, Proc. Amer. Math. Soc. **142** (2014), no. 6, 2119–2122. doi:10.1090/S0002-9939-2014-11929-0. Author version: arXiv:1112.4181v3.
- [25] H. P. McKean, Jr., and I. M. Singer, *Curvature and the eigenvalues of the Laplacian*, J. Differential Geom. **1** (1967), 43–69. doi:10.4310/jdg/1214427880.
- [26] M. J. Micallef and M. Y. Wang, *Metrics with nonnegative isotropic curvature*, Duke Math. J. **72** (1993), no. 3, 649–672. doi:10.1215/S0012-7094-93-07224-9.
- [27] S. B. Myers, *Riemannian manifolds with positive mean curvature*, Duke Math. J. **8** (1941), no. 2, 401–404. doi:10.1215/S0012-7094-41-00832-3.
- [28] M. Obata, *Certain conditions for a Riemannian manifold to be isometric with a sphere*, J. Math. Soc. Japan **14** (1962), no. 3, 333–340. doi:10.2969/jmsj/01430333.
- [29] B. O’Neill, *The fundamental equations of a submersion*, Michigan Math. J. **13** (1966), no. 4, 459–469. doi:10.1307/mmj/1028999604.
- [30] M.-F. Roy, *Certificates of Positivity in the Bernstein basis*, Oberwolfach Reports, report 14/2007, 800–804. Report on joint work with F. Boudaoud, F. Caruso, and R. Leroy. Report.
- [31] J. L. Synge, *On the Connectivity of Spaces of Positive Curvature*, Quart. J. Math. Oxford Ser. **7** (1936), 316–320. doi:10.1093/qmath/os-7.1.316.
- [32] D. Yang, *Rigidity of Einstein 4-manifolds with positive curvature*, Invent. Math. **142** (2000), no. 2, 435–450. doi:10.1007/PL00005792.